



Hybrid-Ensemble Deep-Learning Models to Enhance the Sunspot Prediction and Forecasting of Solar Cycle 26

Aman Kumar¹ · Vipin Kumar²

Received: 25 February 2025 / Accepted: 22 June 2025 / Published online: 17 July 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

This study comprehensively evaluates deep-learning models across three types of sunspot data: 13-month smoothed Sunspot Number (SSN), yearly SSN, and mean monthly SSN data. Traditional models such as LSTM, GRU, CNN, RNN, and BiLSTM are compared against proposed hybrid models: Hybrid1 (CNN-DilatedLSTM-BiLSTM-GRU), Hybrid2 (CNN-GRU-RNN with Dropout Regularization), Hybrid3 (CNN-GRU), Hybrid4 (CNN-GRU-RNN without Dropout) (Hybrid2 and Hybrid4 both integrate CNN-GRU-RNN architectures, but Hybrid2 introduces Dropout layers to reduce overfitting, making it a regularized version of Hybrid4), and a Hybrid Ensemble model (Hybrid2 + Hybrid4). Key metrics like RMSE, MAE, MSE, and R^2 are used to assess model performance. The results indicate that hybrid models consistently outperform traditional models across all datasets. Specifically, the Hybrid Ensemble achieves enhanced predictive accuracy, recording an RMSE of 4.062 and R^2 of 0.9964 for 13-month smoothed SSN data, an RMSE of 22.11 and R^2 of 0.8920 for yearly SSN data, and an RMSE of 24.61 and R^2 of 0.8826 for mean monthly SSN data. These findings demonstrate the ability of hybrid models, especially the Hybrid Ensemble, to effectively capture complex patterns in sunspot time-series data.

In addition to model evaluation, this study provides forecasted SSN values for Solar Cycle 26, projecting a gradual increase in solar activity from 2025 (SSN: 112.39) to a peak in 2036 (SSN: 165.35), followed by a slight decline in 2037 (SSN: 155.25), with the lowest SSN occurring in 2032 (SSN: 10.41). These forecasts align well with known solar cycle variations and offer valuable insights into upcoming solar activity and its implications for space weather, climate, and technology. A Friedman non-parametric test was conducted to rank model performance, confirming the Hybrid Ensemble as the top performer. Holm-adjusted multiple comparisons showed negligible differences, reinforcing the robustness of the hybrid and ensemble approaches. This research highlights the value of combining different architectures to improve forecasting accuracy, especially for complex scientific time-series data such as solar activity.

Keywords Sunspot · Solar cycle · Deep learning · Hybrid model · Ensemble hybrid model · Stack model · Time series forecasting

1. Introduction

The study of sunspots and their influence on solar activity has been a focus area for researchers in space for a long time. Sunspots are visible abnormalities on the sun's outer

surface that provide considerable data about the repetitive example of sun-oriented action as per the study of Zhu, Zhu, and He (2022). The sunspot number (SSN) reflects how often these spots appear. For a myriad of practical applications, such as satellite communications, environmental studies, and space research, understanding and predicting SSN patterns is critical as per the study of Abdulkadir et al. (2023). In the past decades, there have been a lot of logical examinations of SSNs to increase their precision in predicting sunspot number projections and improve our understanding of the solar cycle. Different methods have been applied in such research studies, from complex deep-learning strategies like CNN and LSTM architectures to simpler statistical models like ARIMA and ARMA. This review provides a comprehensive assessment and analysis of the latest advancements in SSN prediction models as per the study of Prasad et al. (2023). We think about the presentation of different DL and non-DL techniques for forecasting SSN patterns, considering lengths such as R^2 , RMSE, MAPE, MAE, and MSE. For further enhancement of expectation precision, we also examine the usage of crossover models and troupe techniques that encompass both DL and conventional methods as per the study of Dhuri et al. (2022), Chung et al. (2014).

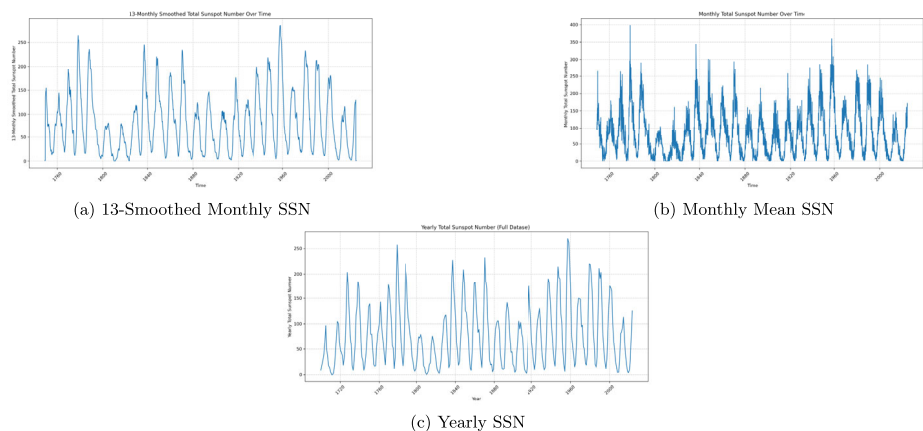
The dataset used in this study is obtained from the Royal Observatory of Belgium's World Data Center SILSO (WDC-SILSO), which provides publicly accessible sunspot number records from the mid-18th century to the present. The data includes monthly, yearly, and daily sunspot numbers. Prior to model training, we applied preprocessing techniques such as missing value handling, anomaly detection, and normalization. In accordance with WDC-SILSO's guidelines, proper citation is provided: 'Source: WDC-SILSO, Royal Observatory of Belgium, Brussels' (www.sidc.be/SILSO/datafiles) WDC-SILSO, Royal Observatory of Belgium (2025). Our proposed strategy incorporates occasional disintegration examination to remove the pattern, occasional, and lingering parts from the SSN information. These broken-down parts are then used to fabricate profound learning models, which recognize the central examples of sun-powered action as per the study of Chola and Benifa (2022), Jiang et al. (2018).

The last estimate is then delivered by recombining the singular projections utilizing a gathering approach. Our review's discoveries show that the recommended model outflanks traditional DL and measurable models in various assessment boundaries. Specifically, the proposed model shows empowering exactness in anticipating SSN patterns for different dataset recurrence varieties, like everyday SSN, month-to-month mean SSN, and month-to-month absolute SSN as per the study of Li et al. (2021, 2018). To summarize, this study adds to the ongoing undertakings to further develop SSN expectation strategies by giving savvy data about utilizing outfit approaches and DL methods in sunlight-based action gauging. The examination depicted here has ramifications for environmental display, space investigation, and regions that rely upon exact sun-powered movement conjectures as per the study of Jiang et al. (2018).

In Figure 1 (a), the smoothed-over 13-month sunspot number (SSN) monthly data graph provides an averaged sunspot activity evolution. The storyline unveils the long-term patterns in sunspot activity. It efficiently removes short-time variability and interference since it's an even curve to get the periodic pattern of solar activity with its distinctive high and low points relating to the approximately 11-year solar cycle. This storyline helps to identify long-term trends and patterns in solar activity as per the study of Su et al. (2024). In Figure 1 (b), the monthly mean SSN data is depicted in a line plot showing each month's recorded average sunspot numbers. The plot gives much more detail about the sunspot activity, with seasonality and month-to-month variation appearing more quickly than in the smoothed data. The storyline frequently exhibits regular highs and lows, similar to how the solar cycle impacts the sunspots' count. It provides insight into cyclic variations and the

Table 1 A compilation of abbreviations along with their complete forms.

Abbreviation	Full form	Abbreviation	Full form
GPU	Graphics Processing Unit	RMSE	Root Mean Square Error
R^2	Coefficient of Determination	MAPE	Mean Absolute Percentage Error
BiLSTM	Bidirectional Long Short-Term Memory	SC26	Solar Cycle 26
CNN	Convolutional Neural Network	KPSS	Kwiatkowski–Phillips–Schmidt–Shin
SSN	Sunspot Number	AIC	Akaike Information Criterion
ReLU	Rectified Linear Unit	SC25	Solar Cycle 25
GRU	Gated Recurrent Unit	MSE	Mean Squared Error
DL	Deep Learning	LSTM	Long Short-Term Memory
ACF	Autocorrelation Function	MAE	Mean Absolute Error
BIC	Bayesian Information Criterion	RNN	Recurrent Neural Network

**Figure 1** Line plot of univariate sunspot data.

overall trend of sunspots during the range of time under study as per the study of Wang, Li, and Guo (2021a). In Figure 1 (c), the yearly SSN data is represented on a line plot showing the average year sunspot numbers. This storyline provides a broader view of sunspot activity by removing monthly and seasonal changes to reveal long-term patterns. It illustrates the general trend of sunspot activity over several years, showing the rise and fall associated with the solar cycle. This plot is useful for examining long-term patterns and the general trend of solar activity. Ultimately, the study of sunspots and their influence on solar activity remains an important area of research with far-reaching implications for space science and technology. Using advanced deep-learning models and a combination of approaches, this work improves our understanding and prediction of sunspot patterns, overcoming key challenges in predicting solar activity as per the study of Kakad, Kakad, and Ramesh (2017). Combining seasonal decomposition and ensemble methods with traditional and modern prediction techniques provides a complete framework to enhance accuracy across different periods. Results emphasize the importance of improved forecasting techniques in enhancing satellite communications, environmental monitoring, and space exploration. This study provides

important knowledge regarding the forecasting of sunspots, as well as opening ways toward future progress in predicting solar activity and the need for continuous innovation and exploration in this dynamic field.

In this study, the author has proposed four hybrid models, i.e., Hybrid Model 1 (CNN-DilatedLSTM-BiLSTM-GRU), Hybrid Model 2 (CNN-GRU-RNN), Hybrid Model 3 (CNN-GRU), and Hybrid Model 4 (CNN-GRU-RNN). A final ensemble model has also been proposed using Hybrid Model 2 and Hybrid Model 4. These proposed models have performed better than the traditional models over the various measures like RMSE, MAPE, MAE, MSE, and R^2 , where the SSN three datasets are used to forecast the sunspot numbers. The models are validated using graphical representation (for predictive pattern) and non-parameter statistical tests (for validation purposes), which also show the effectiveness of the proposed models. Lastly, the solar cycle of 26 years is also forecasted, and the peak number of sunspots is found. And also compared with literature values. Table-1 have compilation of abbreviations along with their complete full form.

Key Contributions of the Research:

- The multiple hybrid models have been identified, i.e., Hybrid Model 1 (CNN-Dilated LSTM-BiLSTM-GRU), Hybrid Model 2 (CNN-GRU-RNN), Hybrid Model 3 (CNN-GRU), and Hybrid Model 4 (CNN-GRU-RNN) with a combination of traditional models of deep learning.
- A ensemble of Hybrid Model 2 and Hybrid Model 4 has also been proposed as a final model along with hybrid models
- The analysis of the proposed models has been performed over the monthly, monthly mean and yearly SSN dataset over the various measures such as RMSE, MAPE, MAE, MSE, and R^2 .
- Further, the Solar Cycle of 26 years has also been forecasted with sunspot values and a comparison of the models has also been done with state-of-the-art models.

2. Literature Review

The current state-of-the-art solar activity prediction has been systematically analyzed through a comprehensive review of existing methodologies. As presented in Table 2, this analysis encompasses three primary categories: statistical models, machine learning approaches including deep learning architectures, and identified research gaps with particular emphasis on learning models and ensemble methods. The review examines the evolution of prediction methodologies from traditional statistical approaches to advanced machine learning implementations. The analysis mainly focuses on transitioning from conventional statistical methods to sophisticated deep learning architectures, highlighting the increasing adoption of hybrid and ensemble approaches. The research gaps identified through this systematic review reveal significant opportunities for advancement, particularly in integrating multiple methodological approaches and developing more robust ensemble techniques.

2.1. Statistical Model

Statistical approaches have played a fundamental role in solar activity forecasting. Early research by Pesnell (2016) employed the Solar Cycle Data Assimilation (SODA) method, which primarily relied on statistical patterns observed in historical solar cycles. While effective for long-term predictions, this approach was limited in capturing short-term variations.

Table 2 Systematic Analysis of Solar Activity Prediction Methods Categorized by Modeling Approaches.

Authors & year	Time period	Methodology	Data resolution	Sample size	Key limitations
Statistical Methods					
Pesnell (2016)	–	SODA	Solar Cycle 25	–	Limited to statistical methods; Lacks deep-learning integration
Han and Yin (2019)	1749 – 2017	Decline Phase Model	Monthly, Smoothed Monthly	–	Focus on basic statistical modeling; Absence of ML techniques
Labonville, Charbonneau, and Lemerle (2019)	–	Solar Cycle Dynamo Model	13-Monthly	–	Traditional modeling approach; Limited predictive capabilities
Petrovay (2020)	–	SFT Modeling	Cycle 25 Forecast	–	Traditional approach; Lacks ML integration
Deep-Learning Approaches					
Zhu, Zhu, and He (2022)	1954 – 2020	Optimized LSTM	13-Monthly, Monthly Mean	789	Single architecture focus; Limited temporal scope
Waykar, Kale, and Karlekar (2022)	1749 – 2022	Deep LSTM	Solar cycle progression	–	Single architecture; Lacks hybrid integration
Prasad et al. (2022)	1750 – 2019	LSTM	Monthly	3240	Basic implementation; No hybrid architectures
Jiang et al. (2023)	1700-present	LSTM	Yearly, Monthly Mean	318	Single architecture; Limited temporal resolution
Hybrid Models					
Lee (2020)	–	EMD-LSTM Hybrid	Solar Cycles 24 – 25	–	Limited temporal scope; Insufficient ensemble integration
Hasoon and Al-Hashimi (2022)	1749-present	DNN-LSTM	Multi-scale	3265	Basic hybrid architecture; Limited decomposition analysis
Kumar et al. (2023)	1700 – 2008	SARMA-LSTM	Annual	309	Limited temporal scope; Basic hybrid implementation

Table 2 (Continued)

Authors & year	Time period	Methodology	Data resolution	Sample size	Key limitations
Kumar and Kumar (2023b)	1749 – 2023	sConvLSTM	Multiple Resolutions	74,906	Limited ensemble integration; Basic architecture
Moustafa and Khodairy (2023)	1749 – 2022	ARIMA-BiLSTM	Monthly Mean, Annual	3285	Basic hybrid approach; Limited decomposition analysis
Kumar and Kumar (2024)	1749 – 2023	CNN-BiGRU	Multi-resolution	75,049	Limited ensemble exploration; Basic hybrid approach
Yang et al. (2024)	1811 – 2023	CEEMDAN-GRU	Monthly Mean	2547	Single resolution focus; Limited hybrid exploration
Ensemble Approaches					
Benson et al. (2020)	1749 – 2019	WaveNet-LSTM	13-Monthly, Monthly Mean	3251, 1744	Limited resolution range; Basic hybrid implementation
Panigrahi et al. (2021)	1749 – 2019	PHM-MAX-ARIMA-ETS-SVM	Monthly, Yearly	3252, 320	Limited to statistical methods; No deep learning integration
Nghiem et al. (2022b)	1749 – 2022	CNN-LSTM	Monthly	3278	Single resolution focus; Limited ensemble integration
Dang et al. (2022)	–	Linear Regression XGBoost-DL	13-month smoothed	–	Basic ensemble approach; Limited temporal resolution
Hu and Rodriguez (2023)	1749 – 2021	CNN-LSTM	Multiple Resolutions	3282	Limited ensemble integration; Basic hybrid architecture
Xu, Jain, and Xing (2024)	1749 – 2023	SARIMA-RF	Multiple Resolutions	–	Basic ensemble approach; Limited ML integration

Using statistical methodologies, as per the study of Han and Yin (2019) introduced the Decline Phase Model using monthly and smoothed monthly sunspot numbers from 1749–2017 and 1933–2017. Their work demonstrated the value of extended-term averages, though it faced challenges in representing the intricate dynamics of solar cycles. As per the study of Panigrahi et al. (2021) made significant contributions by implementing a comprehensive suite of statistical models, including PHM-MAX, ARIMA, ETS, and SVM for solar activity prediction. Their data analysis from 1749 to 2019 provided valuable insights into monthly and yearly patterns. A related study further explored the integration of ARIMA, ETS, and SVM models, demonstrating improved predictive capabilities for long-term solar cycle forecasting. Recent advancements have seen the emergence of hybrid statistical approaches. As per the study of Kumar et al. (2023) developed a SARMA-LSTM hybrid model for annual solar activity prediction using data from 1700 to 2008. This combination of statistical and machine learning techniques showed promise in capturing linear and nonlinear properties of time series data. As per the study of Xu, Jain, and Xing (2024) further advanced the field by integrating SARIMA with Random Forest (RF) for analyzing data from 1749 to 2023. Their approach demonstrated how traditional statistical methods could be enhanced through machine learning techniques, though opportunities remain for incorporating more advanced decomposition methods. These statistical approaches have consistently shown value in solar activity prediction, particularly for long-term forecasting. However, the literature suggests that pure statistical methods may benefit from integrating modern machine learning techniques to capture better the complex, nonlinear patterns inherent in solar activity data.

2.2. Machine Learning and Deep-Learning Model

Deep learning and machine learning models have revolutionized solar activity forecasting through their ability to capture complex patterns and relationships in temporal data. The evolution of these approaches can be traced through several key developments and architectures. The foundation was established with basic neural network architectures, as demonstrated by Pala and Atici (2019), who implemented Neural Network AutoRegressive (NNAR) and Long Short-Term Memory (LSTM) models for analyzing monthly and yearly data. This work provided initial insights into the potential of deep learning for solar forecasting. LSTM networks have proven particularly effective for temporal prediction tasks. As per the study of Zhu, Zhu, and He (2022) developed an optimized LSTM model for 13-monthly and monthly mean data analysis, while Waykar, Kale, and Karlekar (2022), Kumar and Kumar (2023a), implemented a deep LSTM architecture for comprehensive solar cycle progression prediction. As per the study of Prasad et al. (2022) further demonstrated LSTM's capabilities in handling extended time series by analyzing monthly sunspot data spanning nearly 270 years. The field has seen significant advancement through hybrid approaches combining multiple deep-learning techniques. As per the study of Hasoon and Al-Hashimi (2022) integrated Deep Neural Networks with LSTM (DNN-LSTM) to enable multi-scale prediction across various temporal resolutions. As per the study of Nghiem et al. (2022a) developed a CNN-LSTM hybrid that leveraged spatial and temporal pattern recognition capabilities. The latest developments have introduced increasingly complex architectures. As per the study of Kumar and Kumar (2024) proposed a CNN-BiGRU model demonstrating exceptional multi-scale forecasting capability. This approach effectively handled various temporal resolutions from daily to annual predictions. As per the study of Yang et al. (2024) advanced the field further by introducing a CEEMDAN-GRU model, incorporating advanced decomposition techniques with deep learning for enhanced prediction accuracy.

Significant insights have emerged from comparative analyses of different architectures. As per the study of Espuña Fontcuberta et al. (2023) comprehensively evaluated multiple architectures, including MESN, RNN, GRU, and LSTM, for Solar Cycles 22 – 24 prediction. As per the study of Moustafa and Khodairy (2023) explored integrating traditional statistical methods with advanced deep learning through their hybrid ARIMA-BiLSTM approach. The most recent trend in the field has been toward ensemble methods and sophisticated hybrid architectures. As per the study of Kumar and Kumar (2023b) introduced the sConvLSTM architecture for multi-scale temporal prediction, while As per the study of Hu and Rodriguez (2023) demonstrated the effectiveness of CNN-LSTM combinations for comprehensive solar activity forecasting. The evolution of deep learning and machine learning approaches in solar activity prediction demonstrates a clear progression from simple architectures to sophisticated hybrid and ensemble methods. The trend indicates an increasing emphasis on multi-scale analysis capabilities and integrating complementary techniques to capture the complex patterns inherent in solar activity data. This progression has led to significant improvements in prediction accuracy and reliability, though opportunities remain for further advancement by integrating newer deep learning architectures and advanced decomposition methods.

2.3. Research Gaps

2.3.1. Based on Learning Model

While studies like Moustafa and Khodairy (2023) have combined ARIMA with BiLSTM, there remains limited exploration of integrating other traditional statistical models with advanced deep learning architectures. The potential synergy between established statistical methods and emerging deep learning techniques has not been thoroughly investigated, particularly in capturing linear and nonlinear patterns in solar activity data. Current hybrid models demonstrate insufficient incorporation of advanced decomposition methods. Although as per the study of Yang et al. (2024) implemented CEEMDAN-GRU, systematic research does not combine various decomposition techniques (such as VMD, EMD, or EEMD) with different hybrid architectures. This limitation affects the models' ability to analyze multi-scale patterns effectively. The existing research shows limited development of multi-modal hybrid architectures. While Kumar and Kumar (2024) proposed CNN-BiGRU, and Hu and Rodriguez (2023) explored CNN-LSTM combinations, there remains a significant gap in investigating more complex hybrid configurations that could potentially capture different aspects of solar activity patterns simultaneously. Current hybrid models often focus on specific temporal scales. For instance, Kumar et al. (2023) concentrated on annual predictions with SARMA-LSTM, while Nghiem et al. (2022b) focused on monthly data. There is a clear need for hybrid architectures capable of simultaneous effective prediction across multiple temporal resolutions. The research demonstrates a limited exploration of ensemble techniques within hybrid frameworks. While some studies have implemented basic hybrid combinations, there is insufficient investigation into how ensemble methods could enhance hybrid model performance by integrating multiple complementary approaches. These gaps in hybrid modelling research present significant opportunities for advancing the field of solar activity prediction, particularly in developing more sophisticated and comprehensive hybrid architectures that can effectively combine multiple methodological approaches while handling various temporal scales and data types.

Table 3 Statistical Description and Stationarity Analysis of Sunspot Datasets.

Dataset	Freq.	Samples	Period	Std	Max	Min	Mean	25%	50%	75%	KPSS	ADF
13-Smoothed Monthly SNN	Monthly	3306	1749-06- 2024	63.12	285	0	81.78	25.2	71.6	119.17	0.1*	7.05e-18*
Yearly Sunspot	Annual	324	1700–2023	-	269	0	78.52	24.57	65.55	115.47	0.1*	0.0194*
Monthly Mean	Monthly	3306	1749-06- 2024	67.65	398.2	0	81.87	24.12	67.90	122.57	0.1*	5.79e-19*

* Indicates stationarity at 0.05 significance level

2.3.2. Based on Model Ensemble Approach

Benson et al. (2020) implemented a WaveNet-LSTM ensemble, combining wavelet-based analysis with deep learning. However, their approach was limited to 13-monthly and monthly mean data prediction, leaving room to explore higher frequency predictions. As per the study of Dang et al. (2022) explored an ensemble approach combining Linear Regression with XGBoost and deep-learning models. While this integration showed promise for 13-month smoothed monthly data, it did not fully exploit the potential of more advanced ensemble architectures. Xu, Jain, and Xing (2024) developed an ensemble combining SARIMA with Random Forest, demonstrating how traditional statistical methods could be enhanced through machine learning ensembles. However, their work did not incorporate deep learning components that could improve prediction accuracy. Most studies have focused on simple two-model ensembles rather than exploring more complex combinations. For instance, while Kumar and Kumar (2023b) introduced stacked architectures with sConvLSTM, they did not investigate broader ensemble configurations incorporating multiple complementary models. There is limited exploration of dynamic ensemble weighting schemes. Studies like Panigrahi et al. (2021) implemented basic model combinations but did not investigate adaptive weighting strategies that could optimize ensemble performance based on different prediction horizons or data characteristics. The integration of decomposition techniques within ensemble frameworks remains largely unexplored. While Yang et al. (2024) incorporated CEEMDAN in their approach, comprehensive ensemble architectures combining multiple decomposition methods with various predictive models are notably absent from the literature.

3. SSN Dataset

3.1. Exploratory Analysis

The dataset collection for this study is sourced from the World Data Center - Sunspot Index and Long-term Solar Observations (SILSO) at the Royal Observatory of Belgium. As presented in Table 3, the data encompasses three distinct temporal resolutions of sunspot observations. The first dataset consists of 13 smoothed Monthly Sunspot Numbers, comprising 3306 samples from 1749 to June 2024. This dataset demonstrates considerable variability with a mean of 81.785 (standard deviation: 63.12) and ranges from 0 to 285 sunspots. The distribution exhibits right-skewed characteristics, with the median (71.6) being lower than the mean. The second dataset presents Yearly Sunspot Data, containing 324 observations from 1700 to 2023. This annual aggregation shows similar distributional characteristics with

a mean of 78.52 and values ranging from 0 to 269. The data maintains comparable central tendencies to the monthly smoothed data, suggesting consistent long-term patterns in solar activity. The third dataset comprises Monthly Mean Sunspot Data, containing 3306 samples from 1749 to June 2024. This dataset exhibits the highest variability among the three, with a standard deviation of 67.65 and the most considerable maximum value of 398.20. The mean value of 81.87 and median of 67.90 indicate a right-skewed distribution similar to the other datasets. SSN time series decompositions are shown in Figure 2 for 13-Smooth monthly SSN, Yearly, SSN and Monthly SSN datasets.

3.2. KPSS and ADF Statistical Test Analysis

The stationarity analysis of the sunspot datasets, as presented in Table 3, demonstrates consistent results across both the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) and the Augmented Dickey-Fuller (ADF) tests at a 0.05 significance level. The KPSS test evaluates the null hypothesis that a time series is stationary around a mean or deterministic trend. In contrast, the ADF test assesses the null hypothesis that a unit root is present, implying non-stationarity. Thus, a significant ADF result (low P-value) combined with a non-significant KPSS result (high P-value) confirms stationarity.

For the 13-smoothed monthly sunspot data, the KPSS test yielded a P-value of 0.1, supporting the null hypothesis of stationarity. The ADF test produced a highly significant P-value of 7.0548e-18, enabling the rejection of the non-stationarity null hypothesis. This dual confirmation establishes the dataset's suitability for time series analysis without requiring differencing. The Yearly Sunspot Data similarly exhibited stationarity in both tests. The KPSS test reported a P-value of 0.1, while the ADF test showed a P-value of 0.0194, confirming stationarity at the 0.05 significance level. The Monthly Mean Sunspot Data followed the same pattern, with the KPSS test returning a P-value of 0.1 and the ADF test producing a highly significant P-value of 5.7942e-19.

These results confirm the stationarity of all three datasets, as indicated by uniform KPSS P-values of 0.1 and significantly low ADF P-values. This characteristic indicates the absence of unit roots and demonstrates stable statistical properties over time. Such stability is particularly valuable for time series analysis and forecasting, enabling the direct application of various modeling methods without the need for differencing or transformation.

4. Methodology

The flowchart of the proposed model framework is shown in Figure 3. Let us consider the X is the input SSN time series undergoes preprocessing to have standardization as shown in Equation 1:

$$X_{\text{std}} = \frac{X - \mu}{\sigma} \quad (1)$$

where μ is mean and σ is standard deviation. The standardized SSN X_{std} has been split into three sets as Training data: X_{train} , Validation data: X_{val} and Test data: X_{test} . Now the learning model $\mathbb{F}(X_{\text{train}})$ learns from the training data, and validation data X_{val} is used for hyperparameters tuning before selecting the model. Then forecasting of test dataset may be represented as $\hat{Y}$, i.e. $\hat{Y} \rightarrow \mathbb{F}(X_{\text{train}})$. The traditional deep-learning models and proposed hybrid models are described in the subsections below.

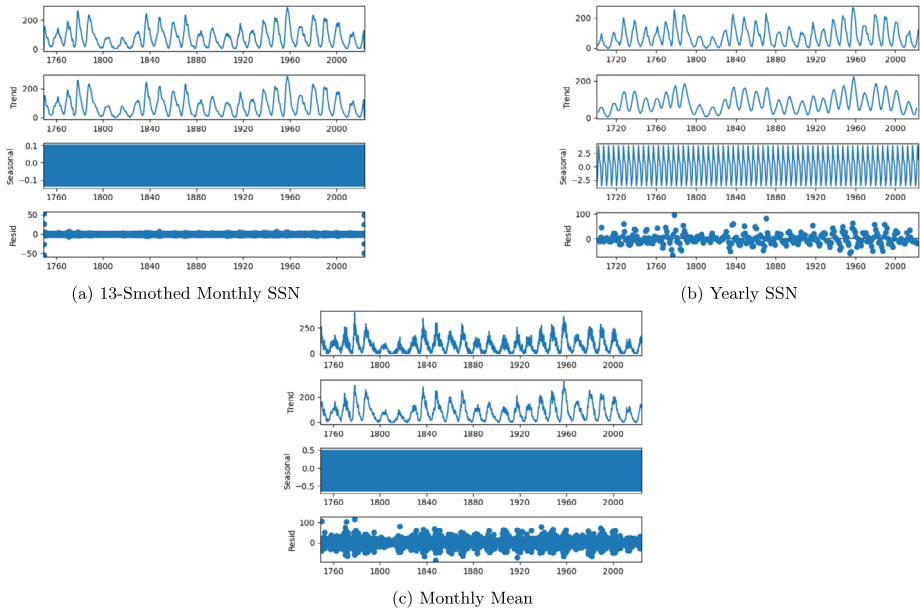


Figure 2 Decomposition of sunspot data into various SSN data: (a) 13-Smoothed Monthly SSN, (b) Yearly SSN, (c) Monthly Mean.

4.1. Traditional Time-Series Deep-Learning Models Description

The traditional deep-learning models are widely used to forecast the SSN, i.e., LSTM, BiLSTM, GRU, CNN, and RNN. The basic details of these models are given:

- LSTM:** LSTM networks are RNNs designed to tackle the issues of exploding and vanishing gradients, ensuring the information present in sequence data is stored when passed through these units. The implementations based on LSTMs result in high efficiency for tasks involving sequential data since they can easily detect correlations between the current and previous sequences. These units are handy for various operations, including language translation, speech recognition, and prediction of future values. LSTMs increase forecasting and translations incrementally by tapping into their ability to capture extended relationships in data.

$$\begin{aligned}
 \phi_t &= \sigma(\Omega_\phi \cdot [\zeta_{t-1}, \delta_t] + \beta_\phi) \\
 \iota_t &= \sigma(\Omega_\iota \cdot [\zeta_{t-1}, \delta_t] + \beta_\iota) \\
 \tilde{\mu}_t &= \tanh(\Omega_\mu \cdot [\zeta_{t-1}, \delta_t] + \beta_\mu) \\
 \mu_t &= \phi_t \odot \mu_{t-1} + \iota_t \odot \tilde{\mu}_t \\
 \omega_t &= \sigma(\Omega_\omega \cdot [\zeta_{t-1}, \delta_t] + \beta_\omega) \\
 \zeta_t &= \omega_t \odot \tanh(\mu_t)
 \end{aligned} \tag{2}$$

- BiLSTM:** BiLSTM networks have two LSTM layers. The first does it in the forward direction, and the second in the reverse direction while observing the sequence. The two-

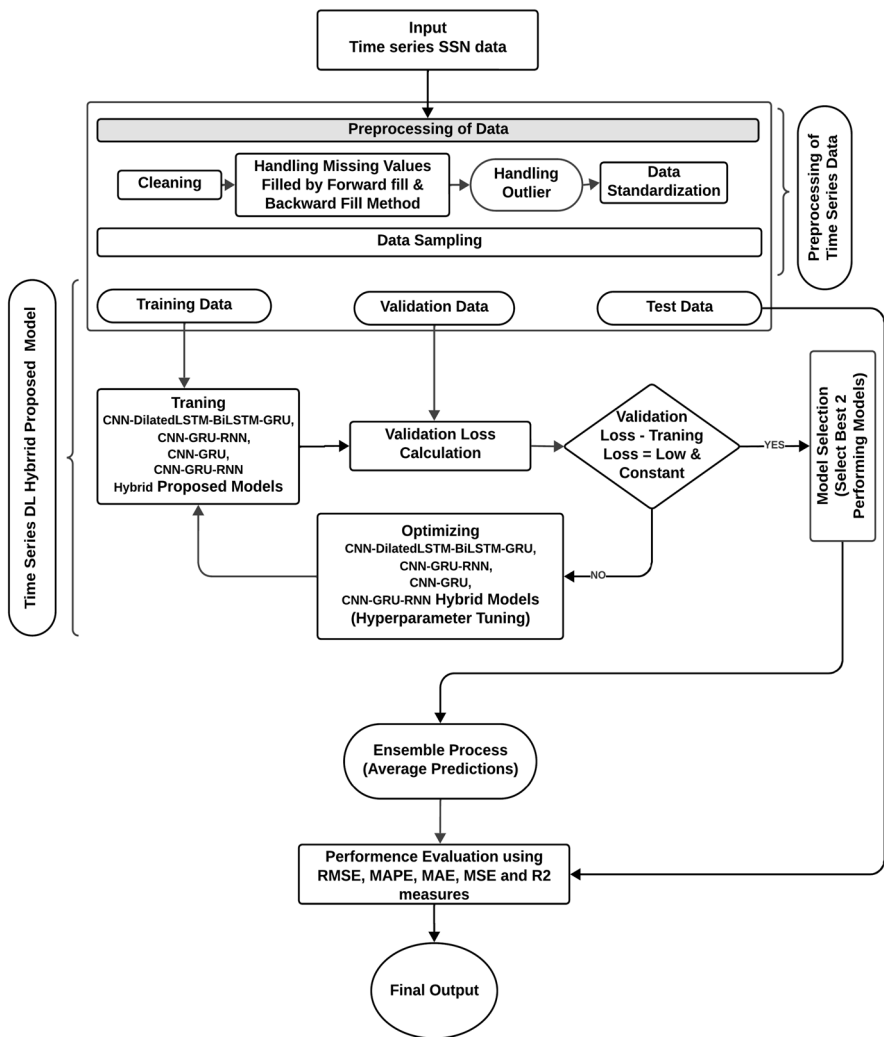


Figure 3 The framework of the proposed Hybrid models.

layered architecture makes it easier to understand interrelationships in the sequence. BiLSTM improves performance on tasks such as language understanding or prediction of future value by considering information from both directions to understand context and relationships between data better.

$$\begin{aligned}
 \vec{\zeta}_t &= \text{LSTM}(\delta_t, \vec{\zeta}_{t-1}) \\
 \overleftarrow{\zeta}_t &= \text{LSTM}(\delta_t, \overleftarrow{\zeta}_{t+1}) \\
 \zeta_t &= [\vec{\zeta}_t; \overleftarrow{\zeta}_t]
 \end{aligned} \tag{3}$$

- **GRU:** The GRU is a variation of RNN that provides a more straightforward alternative than LSTM networks. GRUs are excellent at handling sequential data and, most importantly, mastering distant relationships with the data at hand. These two gates apply: the reset gate and the update gate. The update gate aids in deciding the amount of previous information it needs to memorize for that particular step by helping in deciding which past points are essential. The reset gate determines the degree to which the current input should replace the previous data and helps ignore insignificant past events. This gating mechanism makes GRUs capable of effectively controlling the information flow within the network.

$$\begin{aligned}
 \rho_t &= \sigma(\Omega_\rho \cdot [\zeta_{t-1}, \delta_t] + \beta_\rho) \\
 \nu_t &= \sigma(\Omega_\nu \cdot [\zeta_{t-1}, \delta_t] + \beta_\nu) \\
 \tilde{\zeta}_t &= \tanh(\Omega \cdot [\rho_t \odot \zeta_{t-1}, \delta_t] + \beta) \\
 \zeta_t &= (1 - \nu_t) \odot \zeta_{t-1} + \nu_t \odot \tilde{\zeta}_t
 \end{aligned} \tag{4}$$

- **CNN:** CNNs is the category of design in the neural network field used for deep learning with images and video, which uses convolutions. They are layers of small sections taken from input images and evaluated one at a time by small filters to identify edge characteristics and texture, among other key features. Their ability to learn from their mistakes and being extremely flexible makes CNNs exceptionally adept at mastering the tasks involving the analysis of images and videos.

For 1D CNN with ϕ filters and kernel size κ :

$$\zeta_t^{(l)} = \sigma \left(\sum_{i=1}^{\kappa} \omega_i^{(l)} \cdot \delta_{t+i-1}^{(l-1)} + \beta^{(l)} \right) \tag{5}$$

- **RNN:** The neural networks that can utilize the output of a one-time step as input to the next time step, thereby being ideally suited for the sequential data type, are referred to as recurrent neural networks or RNNs. RNNs process information incrementally as it flows, which allows for remembering past inputs using the present and past inputs. RNNs have applications in tasks with temporal dependencies, such as language modelling or predicting time series.

$$\zeta_t = \sigma(\Omega \cdot [\zeta_{t-1}, \delta_t] + \beta) \tag{6}$$

4.2. Description of Multiple Proposed Hybrid Models and Ensemble Models

4.2.1. Proposed Hybrid 1 Model: CNN-DilatedLSTM-BiLSTM-GRU

In Figure 4, the Hybrid1 model proposed for predicting sunspots combines various deep learning frameworks to leverage each model's capabilities in capturing local patterns and long-term dependencies in the one-dimensional time series data. The process begins with a CNN layer, which extracts local features from the input sequence. This layer employs 64 filters with a 12-unit kernel and applies the ReLU activation function to capture significant spatial patterns in small data segments. The CNN output is flattened to decrease dimensionality, preparing it to merge with other models' outputs. The second component of the hybrid model incorporates the Dilated LSTM architecture. Initially, a convolutional layer with a dilation rate set to 2 is applied to capture patterns occurring at various time scales. Two LSTM

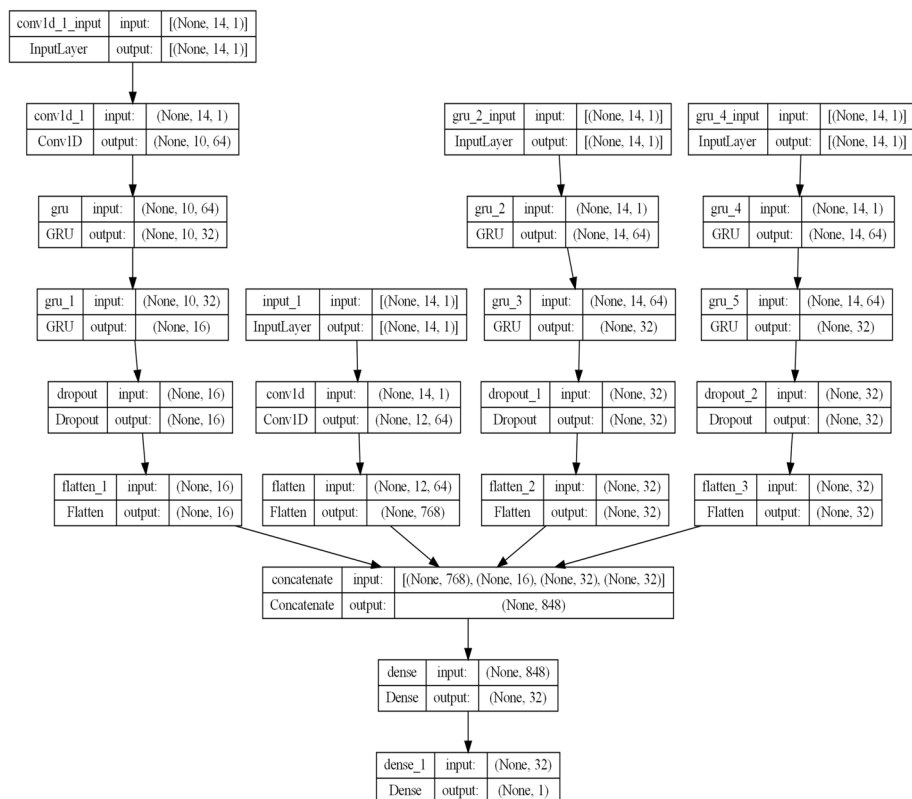


Figure 4 Hybrid1 architecture: CNN-DilatedLSTM-BiLSTM-GRU Hybrid.

layers are used: the first has 32 units and utilizes ReLU as the activation function to maintain the continuity of the model over time, while the second has 16 units and Swish activation to decode long-term dependencies. A dropout layer with a dropout rate of 0.2 follows, and the output is flattened for further processing. The third component utilizes a Bidirectional LSTM (BiLSTM) network, enabling the model to comprehend information from both past and future parts of the sequence. This component consists of two BiLSTM layers that utilize ReLU activation; the first layer contains 64 units, while the second has 32 units. A dropout layer with a dropout rate 0.2 ensures regularization, and the output is flattened. The final component of the hybrid architecture is a GRU network consisting of two GRU layers: the first has 64 units, and the second has 32 units, both applying the ReLU activation function. GRU networks efficiently learn long-term dependencies and overcome the vanishing gradient problem, making them well-suited for time-series data analysis. The result is flattened after applying dropout to prevent overfitting. Subsequently, the results from the four different structures—CNN, Dilated LSTM, BiLSTM, and GRU—are combined and passed through a dense layer consisting of 32 units that utilize ReLU as the activation function to enhance the acquired representations. Finally, a Dense layer with a single neuron generates the sunspot value prediction. The model is compiled using the Adam optimizer, and the MSE is taken as the loss function, aiming to minimize prediction errors and enhance performance. This hybrid structure effectively combines the strengths of CNN layers for feature extraction with recurrent layers (LSTM, BiLSTM, and GRU) for recognizing temporal patterns of SSN, re-

Table 4 Branch-wise Description of Proposed Hybrid 1 Model: CNN-DilatedLSTM-BiLSTM-GRU Architecture.

Branch	Architecture	Mathematical formulation	Description
Branch 1	CNN-GRU Path with Dropout • Conv1D (64 filters) • GRU (32 units) • GRU (16 units) • Dropout • Flatten	$\text{CNN}_1(\Delta) = \sigma(\omega_c * \Delta + b_c)$ $\text{GRU}_1(\xi) \rightarrow \zeta_1 \in \mathbb{R}^{10 \times 32}$ $\text{GRU}_2(\zeta_1) \rightarrow \zeta_2 \in \mathbb{R}^{16}$ $\Xi_1 = \text{Flatten}(\text{Dropout}(\zeta_2))$	Extracts local features using CNN and processes temporal dependencies through dual GRU layers. Dropout prevents overfitting. Output dimension: 16
Branch 2	Conv1D Path • Conv1D (64 filters) • Flatten	$\text{Conv1D}(\Delta) = \sigma(\omega_{c1} * \Delta + b_{c1})$ $\Xi_2 = \text{Flatten}(\text{Conv1D}(\Delta))$	Focuses on local pattern extraction using convolutional operations. Output dimension: 768
Branch 3	GRU Path 1 • GRU (64 units) • GRU (32 units) • Dropout • Flatten	$\text{GRU}_3(\Delta) \rightarrow \eta_1 \in \mathbb{R}^{14 \times 64}$ $\text{GRU}_4(\eta_1) \rightarrow \eta_2 \in \mathbb{R}^{32}$ $\Xi_3 = \text{Flatten}(\text{Dropout}(\eta_2))$	Processes temporal patterns with dual GRU layers of decreasing units. Dropout for regularization. Output dimension: 32
Branch 4	GRU Path 2 • GRU (64 units) • GRU (32 units) • Dropout • Flatten	$\text{GRU}_5(\Delta) \rightarrow \theta_1 \in \mathbb{R}^{14 \times 64}$ $\text{GRU}_6(\theta_1) \rightarrow \theta_2 \in \mathbb{R}^{32}$ $\Xi_4 = \text{Flatten}(\text{Dropout}(\theta_2))$	Parallel GRU path for additional temporal feature extraction. Similar structure to Branch 3. Output dimension: 32
Merged	Dense Layers • Concatenate • Dense (32 units) • Dense (1 unit)	$\Xi_{\text{merged}} = [\Xi_1; \Xi_2; \Xi_3; \Xi_4]$ $\text{Dense}_1 = \sigma(\Omega_1 \cdot \Xi_{\text{merged}} + \beta_1)$ $\hat{Y} = \Omega_2 \cdot \text{Dense}_1 + \beta_2$	Combines features from all branches through concatenation and processes through dense layers for final prediction. Output dimension: 1

where Δ is input data, σ is ReLU activation, $*$ denotes convolution, $[\cdot; \cdot]$ is concatenation, and ω , Ω , b , β are weights and biases respectively

sulting in an efficient model for sunspot forecasting. The mathematical description of the method is shown in Table 4.

4.3. Proposed Hybrid 2 Model: CNN-GRU-RNN

In Figure 5, the hybrid2 model created for forecasting sunspots incorporates various neural network structures, such as convolutional layers and GRU layers, to capture patterns in both space and time within the time-series data. The structure successfully merges the ability to extract features from convolutional neural networks (CNN) with the sequence modelling power of gated recurrent units (GRU), resulting in a robust hybrid model. The data, in the format of `(train.shape[1], train.shape[2])`, is initially passed through a CNN module featuring three 1D convolutional layers. The first convolutional layer employs 64 filters with a size of 12 and a dilation rate set to 2 to detect patterns at different time scales. This layer utilizes ReLU as the activation function and applies padding to ensure the

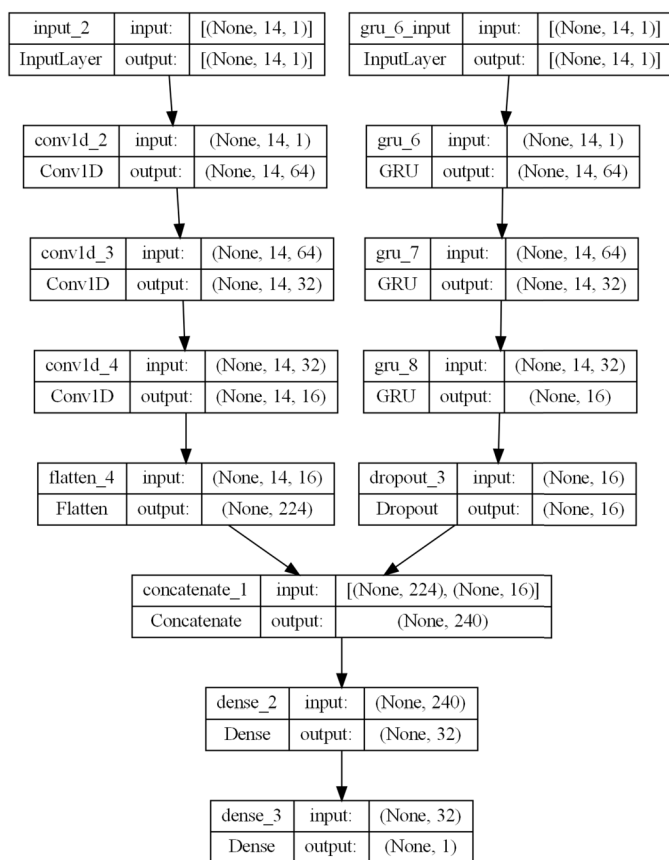


Figure 5 Hybrid2 architecture: CNN-GRU-RNN Hybrid.

output length equals the inputs. The second convolutional layer enhances the features using 32 filters with the same dilation rate and activation. In the third convolutional layer, 16 filters compress the features further. After passing through the layers, the feature maps are flattened to create a feature representation, which will be combined with the outputs of the GRU layer. The GRU network in the model's second part excels at capturing the sequential dependencies present in the sunspot data. This network begins with a 64-unit layer, using the Swish activation function to capture complex temporal relationships and patterns. Finally, two additional GRU layers of 32 and 16 units exist, respectively. It utilizes both Swish and ReLU activation functions to capture short-term as well as long-term relationships. A dropout layer has been added to prevent overfitting by randomly deactivating some neurons in this model part. The sequence data is further processed by three Simple RNN layers in a parallel path, with the layers becoming progressively simpler: the first has 32 units, then decreases to 16, and finally, eight units in the third. ReLU is the activation function for the neurons, gradually building and summarizing temporal characteristics. These layers include another dropout layer with the same dropout rate to bolster generalization. The CNN and GRU results are merged with a concatenation mechanism that combines the CNN's spatial features with the GRU network's temporal patterns. The combined result is then fed into a fully connected dense layer with 32 units and ReLU activation to enhance the merged

Table 5 Path-wise Description of Proposed Hybrid 2 Model: CNN-GRU-RNN Architecture.

Path	Architecture	Mathematical formulation	Description
Path 1	Multi-Layer CNN • Conv1D (64 filters) • Conv1D (32 filters) • Conv1D (16 filters) • Flatten	$\text{Conv1D}_1(\Delta) = \sigma(\omega_1 *_d \Delta + b_1)$ $\text{Conv1D}_2(\xi_1) = \sigma(\omega_2 *_d \xi_1 + b_2)$ $\text{Conv1D}_3(\xi_2) = \sigma(\omega_3 *_d \xi_2 + b_3)$ $\Xi_{\text{CNN}} = \text{Flatten}(\text{Conv1D}_3(\xi_2))$	Three 1D convolutional layers with dilation rate 2, decreasing filter sizes (64 → 32 → 16). Uses ReLU activation and processes spatial features. Output dimension: 224
Path 2	GRU Network • GRU (64 units) • GRU (32 units) • GRU (16 units) • Dropout	$\text{GRU}_1(\Delta) \rightarrow \eta_1 \in \mathbb{R}^{14 \times 64}$ $\text{GRU}_2(\eta_1) \rightarrow \eta_2 \in \mathbb{R}^{14 \times 32}$ $\text{GRU}_3(\eta_2) \rightarrow \eta_3 \in \mathbb{R}^{16}$ $\Xi_{\text{GRU}} = \text{Dropout}(\eta_3)$	Three GRU layers with decreasing units (64 → 32 → 16). It uses Swish and ReLU activations to capture temporal patterns. Includes dropout for regularization. Output dimension: 16
Merged	Dense Layers • Concatenate • Dense (32 units) • Dense (1 unit)	$\Xi_{\text{merged}} = [\Xi_{\text{CNN}}; \Xi_{\text{GRU}}]$ $\text{Dense}_1 = \sigma(\Omega_1 \cdot \Xi_{\text{merged}} + \beta_1)$ $\hat{Y} = \Omega_2 \cdot \text{Dense}_1 + \beta_2$	Combines CNN spatial features (224) and GRU temporal features (16) through concatenation. Processes through two dense layers for final prediction. Output dimensions: 240 → 32 → 1

Notes: Δ is input data, σ is activation function, $*_d$ is dilated convolution with rate d , $[\cdot; \cdot]$ is concatenation, ω , Ω are weights and b , β are biases. Model uses Adam optimizer with MSE loss: $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$

characteristics further. Only a single neuron dense output layer is used to predict the sunspot number at the end. The model is compiled with the Adam optimizer, which applies real-time learning rate adjustments for faster convergence and reduces prediction error using a mean squared error loss function. Such a hybrid design captures spatial and temporal relationships in sunspot data with higher predictive precision. Table 5 presents a comprehensive step-by-step mathematical formulation of the model and concise descriptions of each component.

4.4. Proposed Hybrid 3 Model: CNN-GRU

In Figure 6, this hybrid3 model approach for forecasting sunspots combines CNN with GRU to encompass spatial and temporal trends in the time-series information. This design utilizes CNNs for feature extraction and GRUs for sequence modelling, resulting in a precise and effective prediction system. The time-series data is initially processed by the CNN layers, with its shape determined by `input_shape`. The CNN module consists of three 1D convolutional layers. In the first layer, 64 filters are applied with a kernel size of 6, and the ReLU activation function is used to detect local patterns and features in the input sequence. The second layer uses 32 filters, and the third uses 16 filters with ReLU activation. “Same” padding is also applied to every layer to maintain the input and output dimensions that preserve the time structure of the data. The outputs from the CNN layers are flattened into a one-dimensional vector, depicting features that identify local patterns in sunspot data. At the same time, this hybrid model incorporates a sequential model based on the GRU cell,

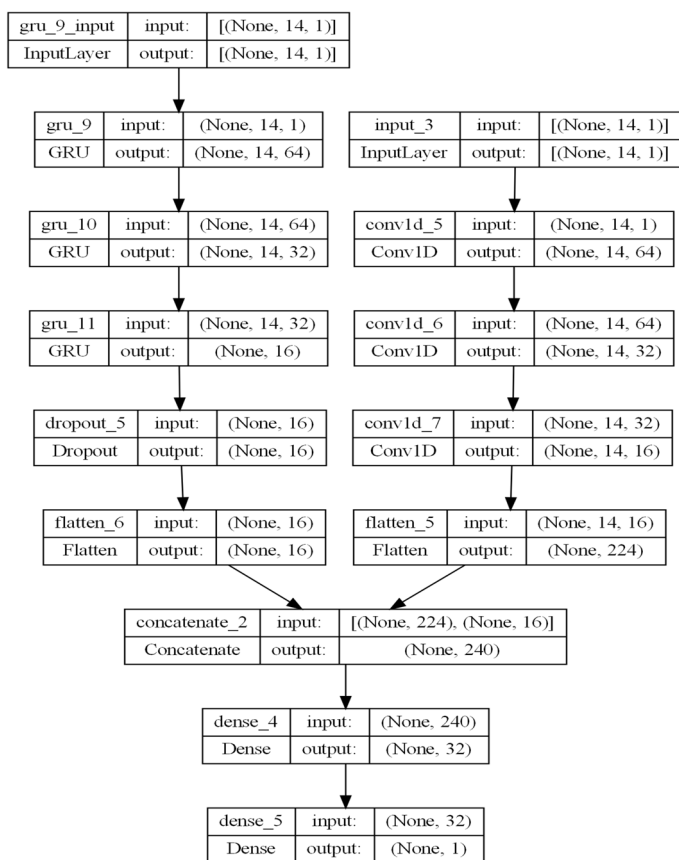


Figure 6 Hybrid3 architecture: CNN-GRU Hybrid.

which should understand the time-related patterns in this time-series data. The GRU network begins with a 64-unit layer and uses the Swish activation function to represent complex patterns as time progresses effectively. This is followed by two additional GRU layers with 32 and 16 units, utilizing a combination of Swish and ReLU activation functions to capture short- and long-term dependencies in the sunspot time series. A Dropout layer is added to avoid overfitting by randomly deactivating a portion of the neurons during training. Then, the output of the GRU is flattened to match the output of the CNN, ensuring that they can be easily merged. The outputs of the CNN layers and the GRU network are combined after extracting spatial features from the CNN layers and temporal features from the GRU network. This combined outcome is passed to a dense layer of 32 units with ReLU activation, further refining the learned feature representations. Finally, a Dense output layer containing one neuron produces the predicted sunspot number. The hybrid model is built with the Adam optimizer, which has a self-tuned learning rate mechanism, and the MSE loss function helps minimize prediction error. This design effectively merges local feature extraction and sequence modelling to improve accuracy in predicting sunspots. Step-wise mathematical Description of this model is shown in Table 6 with short descriptions. Though GRU+CNN hybrids have been utilized in previous literature, the Hybrid 3 model proposed in this work utilizes a tailored parallel architecture, wherein the CNN branch extracts multi-

Table 6 Branch-wise Description of Proposed Hybrid 3 Model: CNN-GRU Architecture.

Branch	Architecture	Mathematical formulation	Description
CNN Path	Three Conv1D Layers • Conv1D (64 filters) • Conv1D (32 filters) • Conv1D (16 filters) • Flatten	$\text{Conv1D}_1(\Delta) = \sigma(\omega_1 * \Delta + b_1)$ $\text{Conv1D}_2(\xi_1) = \sigma(\omega_2 * \xi_1 + b_2)$ $\text{Conv1D}_3(\xi_2) = \sigma(\omega_3 * \xi_2 + b_3)$ $\Xi_{\text{CNN}} = \text{Flatten}(\text{Conv1D}_3(\xi_2))$	Extracts local features using decreasing filter sizes (64 → 32 → 16). Uses ReLU activation and the same padding. Flattened output dimension: 224
GRU Path	Three GRU Layers with Dropout • GRU (64 units, Swish) • GRU (32 units, ReLU) • GRU (16 units, ReLU) • Dropout (0.2) • Flatten	$\text{GRU}_1(\Delta) \rightarrow \eta_1 \in \mathbb{R}^{14 \times 64}$ $\text{GRU}_2(\eta_1) \rightarrow \eta_2 \in \mathbb{R}^{14 \times 32}$ $\text{GRU}_3(\eta_2) \rightarrow \eta_3 \in \mathbb{R}^{16}$ $\Xi_{\text{GRU}} = \text{Flatten}(\text{Dropout}(\eta_3))$	Processes temporal patterns with mixed activations (Swish/ReLU). Decreasing units capture hierarchical temporal features. Dropout prevents overfitting. Output dimension: 16
Merged	Dense Layers • Concatenate • Dense (32 units, ReLU) • Dense (1 unit)	$\Xi_{\text{merged}} = [\Xi_{\text{CNN}}; \Xi_{\text{GRU}}]$ $\text{Dense}_1 = \sigma(\Omega_1 \cdot \Xi_{\text{merged}} + \beta_1)$ $\hat{Y} = \Omega_2 \cdot \text{Dense}_1 + \beta_2$	Combines spatial (CNN) and temporal (GRU) features. Processes through dense layers for final prediction. Uses ReLU activation in a hidden layer. Output dimension: 1

where Δ is input data, σ is ReLU activation, $\text{Swish}(x) = x \cdot \text{sigmoid}(x)$, $*$ denotes convolution, $[\cdot; \cdot]$ is concatenation, and ω , Ω , b , β are weights and biases respectively

level convolutional features and the GRU branch captures long-term dependencies through deep recurrent layers with different activations (swish, relu). The two branches' outputs are combined with a late fusion approach, allowing for a richer representation that suits the solar activity dynamics.

4.5. Proposed Hybrid Model 4: CNN-GRU-RNN

Figure 7, a hybrid4 model aimed at sunspot forecasting by combining three complex deep learning structures, namely CNN, GRU, and RNN, to incorporate spatial and temporal inter-relationships into the one-variable time series data. The model begins with a CNN module featuring three 1D convoluted layers applied to the input. The first layer uses 64 filters of size 12 and applies ReLU to identify local patterns and features. In the second layer, 32 filters are applied, and complexity is reduced using 16 filters in the third layer, with both layers again applying ReLU. Padding is applied uniformly in each layer to ensure that output dimensions are consistent with the input dimensions. The output of the CNN is flattened to create a feature representation that captures local temporal patterns, which is later combined with the output of the GRU network. Simultaneously, the model employs a sequential network based on GRU to understand long-term patterns and connections in the sunspot data. The GRU network consists of three tiers, starting with an initial layer with 64 units, which utilizes the Swish activation function to obtain detailed and nonlinear temporal relationships. The second and third tiers contain 32 and 16 units, combining Swish and ReLU

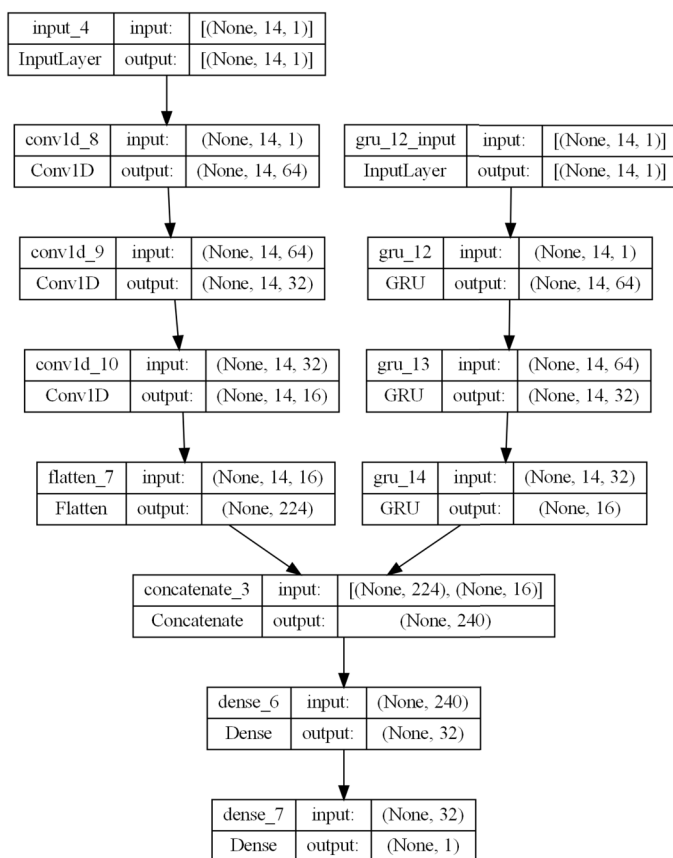


Figure 7 Hybrid4 architecture: CNN-GRU-RNN Hybrid.

activations. The purpose of the GRU network is to effectively grasp temporal relationships by regulating information flow with its gating system, which helps mitigate the problem of vanishing gradient standards in recurrent models. The third part of the hybrid model consists of a sequential RNN network with three SimpleRNN layers. The first layer comprises 32 units, followed by a layer with 16 units, and finally, the last layer consists of 8 units, all activated using ReLU. These RNN layers learn sequential patterns in the data, providing additional temporal information to enhance the model's learning capabilities. The outputs from the CNN and GRU are combined through concatenation, merging the local features of the CNN with the temporal patterns of the GRU. The combined output then undergoes a Dense layer with 32 units, followed by ReLU. Finally, the Dense output layer contains only one neuron, which outputs the sunspot value. With an Adam optimizer that adjusts its learning rate during compilation for faster convergence, MSE loss is applied to assess prediction efficiency. This combination of CNNs for feature extraction and RNN/GRU layers for learning temporal patterns is designed to improve sunspot prediction accuracy at various time scales. Table-7 have the mathematical formulation of Hybrid CNN-GRU-RNN.

Table 7 Branch-wise Description of Proposed Hybrid Model-4: CNN-GRU-RNN Architecture.

Branch	Architecture	Mathematical formulation	Description
CNN Path	Three Conv1D Layers • Conv1D (64 filters) • Conv1D (32 filters) • Conv1D (16 filters) • Flatten	$\text{Conv1D}_1(\Delta) = \sigma(\omega_1 * \Delta + b_1)$ $\in \mathbb{R}^{14 \times 64}$ $\text{Conv1D}_2(\xi_1) = \sigma(\omega_2 * \xi_1 + b_2)$ $\in \mathbb{R}^{14 \times 32}$ $\text{Conv1D}_3(\xi_2) = \sigma(\omega_3 * \xi_2 + b_3)$ $\in \mathbb{R}^{14 \times 16}$ $\Xi_{\text{CNN}} = \text{Flatten}(\text{Conv1D}_3(\xi_2))$	Extracts local features using three convolutional layers with decreasing filter sizes (64 → 32 → 16). Uses a kernel size of 12 and ReLU activation. The same padding maintains temporal dimension. Flattened output dimension: 224
GRU Path	Three GRU Layers • GRU (64 units, Swish) • GRU (32 units, Swish) • GRU (16 units, ReLU)	$\text{GRU}_1(\Delta) \rightarrow \eta_1 \in \mathbb{R}^{14 \times 64}$ $\text{GRU}_2(\eta_1) \rightarrow \eta_2 \in \mathbb{R}^{14 \times 32}$ $\text{GRU}_3(\eta_2) \rightarrow \eta_3 \in \mathbb{R}^{16}$ $\Xi_{\text{GRU}} = \eta_3$	Processes temporal patterns with mixed activations (Swish/ReLU). Uses a gating mechanism to control information flow. Hierarchical structure captures multi-scale temporal dependencies. Output dimension: 16
Merged	Dense Layers • Concatenate • Dense (32 units, ReLU) • Dense (1 unit)	$\Xi_{\text{merged}} = [\Xi_{\text{CNN}}; \Xi_{\text{GRU}}]$ $\in \mathbb{R}^{240}$ $\text{Dense}_1 = \sigma(\Omega_1 \cdot \Xi_{\text{merged}} + \beta_1)$ $\in \mathbb{R}^{32}$ $\hat{Y} = \Omega_2 \cdot \text{Dense}_1 + \beta_2$ $\in \mathbb{R}^1$	Combines spatial features from CNN and temporal features from GRU. Processes through dense layers with ReLU activation. The final layer produces single-value prediction using linear activation. Total parameters optimized using Adam optimizer with MSE loss.

Notation: Δ : Input time series data $\in \mathbb{R}^{14 \times 1}$, $\sigma(\cdot)$: ReLU activation function, $\text{Swish}(x) = x \cdot \text{sigmoid}(x)$, $*$: Convolution operation, $[\cdot; \cdot]$: Concatenation operation, ω_i : Convolutional filter weights, Ω_i : Dense layer weights, b_i , β_i : Bias terms

4.6. Hybrid Ensemble of Hybrid Model 2 and Model 4 to Enhance Sunspot Number Forecasting

The proposed Hybrid Ensemble Model combines two high-performing hybrid architectures - Hybrid Model 2 (CNN-GRU-RNN) and Hybrid Model 4 (CNN-GRU-RNN) - with the help of an averaging ensemble method. The reason behind selecting these models to form an ensemble is their performance in RMSE values over all three SSN datasets. The proposed ensemble approach integrates two sophisticated hybrid architectures (Hybrid2 and Hybrid4) to enhance sunspot forecasting accuracy by leveraging their complementary strengths. Both constituent models incorporate a three-branch architecture combining CNN, GRU, and RNN networks but with distinct implementations that capture different aspects of the temporal patterns. the mathematical configuration of the proposed model is shown in Table-8. Hy-

Table 8 Proposed Hybrid Ensemble of Hybrid Model 2 and Model 4 to Enhance Sunspot Number Forecasting.

Component	Architecture	Mathematical formulation	Description
Hybrid Model 2	CNN-GRU Architecture • CNN Branch • GRU Branch • Dense Layers	$\Xi_{\text{CNN}}^{(2)} = \text{Flatten}(\text{CNN}_2(\Delta))$ $\Xi_{\text{GRU}}^{(2)} = \text{GRU}_2(\Delta)$ $\Xi_{\text{merged}}^{(2)} = [\Xi_{\text{CNN}}^{(2)}; \Xi_{\text{GRU}}^{(2)}]$ $\hat{Y}^{(2)} = \text{Dense}(\Xi_{\text{merged}}^{(2)})$	The first hybrid model combines CNN for spatial features and GRU for temporal patterns. It includes dropout and dense layers for the final prediction. Output dimension: 1
Hybrid Model 4	CNN-GRU Architecture • CNN Branch • GRU Branch • Dense Layers	$\Xi_{\text{CNN}}^{(4)} = \text{Flatten}(\text{CNN}_4(\Delta))$ $\Xi_{\text{GRU}}^{(4)} = \text{GRU}_4(\Delta)$ $\Xi_{\text{merged}}^{(4)} = [\Xi_{\text{CNN}}^{(4)}; \Xi_{\text{GRU}}^{(4)}]$ $\hat{Y}^{(4)} = \text{Dense}(\Xi_{\text{merged}}^{(4)})$	A second hybrid model with similar architecture but different hyperparameters. Independently trained for diversity in predictions. Output dimension: 1
Ensemble	Averaging Method • Model Averaging • MSE Loss	$\hat{Y}_{\text{ensemble}} = \frac{1}{2}(\hat{Y}^{(2)} + \hat{Y}^{(4)})$ $\mathcal{L}_{\text{ensemble}} = \frac{1}{N} \sum (y_i - \hat{Y}_{\text{ensemble},i})^2$	Combines predictions from both models through simple averaging. Takes advantage of the complementary strengths of both models. Final prediction optimized using MSE loss. Output dimension: 1

where: Δ : Input time series data, CNN_i , GRU_i : CNN and GRU networks for model i , $\hat{Y}^{(i)}$: Prediction from model i , $\hat{Y}_{\text{ensemble}}$: Final ensemble prediction, $\mathcal{L}_{\text{ensemble}}$: Ensemble loss function

brid2 employs dilated convolutions with a rate of 2 in its CNN branch, comprising three layers with 64, 32, and 16 filters, respectively, while Hybrid4 utilizes uniform padding in its convolutional layers with the same filter configuration. The GRU branch in both models consists of three layers (64, 32, 16 units), with Hybrid2 implementing Swish activation followed by ReLU and Hybrid4 combining both activations throughout its layers. The RNN branch in both models follows a progressive unit reduction ($32 \rightarrow 16 \rightarrow 8$) with ReLU activation, though Hybrid2 incorporates additional dropout layers for enhanced generalization. The ensemble prediction $\hat{Y}_{\text{ensemble}}$ is computed through a weighted combination $\alpha \hat{Y}_{\text{hybrid2}} + (1 - \alpha) \hat{Y}_{\text{hybrid4}}$, where α is optimized during validation to minimize the ensemble error. This weighted averaging mechanism allows the model to adaptively balance the contributions of each hybrid architecture, potentially leading to more robust and accurate predictions. The entire ensemble framework is trained using the Adam optimizer with Mean Squared Error (MSE) as the loss function, ensuring optimal convergence and prediction accuracy. The complementary nature of the two hybrid models, combined with their weighted integration, enables the ensemble to capture fine-grained temporal patterns and long-term dependencies in sunspot time series data, potentially outperforming individual models in prediction stability and accuracy.

Table 9 Hybrid Model Parameters.

Model	Parameters
Hybrid 1 (CNN-DilatedLSTM-BiLSTM-GRU)	dilation rate= 2, activation='swish', activation='relu', optimizer='adam', loss function='mse'
Hybrid 2 (CNN-GRU-RNN)	activation='relu', dilation rate= 2, activation='swish', optimizer='adam', loss='mse'
Hybrid 3 (CNN-GRU)	activation='relu', activation='swish', optimizer='adam', loss='mse'
Hybrid 4 (CNN-GRU-RNN)	activation='relu', activation='swish', optimizer='adam', loss='mse'

5. Experimental Setup

In Table 9, the suggested hybrid models—Hybrid1, Hybrid2, Hybrid3, Hybrid4, and the Hybrid ensemble model—indicate a substantial improvement over DL learning methods such as LSTM, BiLSTM, GRU, CNN, and RNN. These hybrid models are carefully adjusted with hyperparameters designed to enhance their effectiveness. The study employs a batch size of 66 and the Adam optimizer to progress through the training phase effectively. The models utilize a dilation rate of 2 and activation functions `swish` and `relu` for non-linearity. MSE serves as the loss function to measure prediction accuracy. Each model undergoes extensive training for 250 epochs, with early stopping implemented to prevent overfitting. Early stopping activates if the validation loss does not improve for 20 epochs. The dataset is split balanced, allocating 60% of the data for training and the remaining 40% for validation. This technique ensures that the models are trained on past data and tested on recent data to evaluate their predictive capability. The univariate data containing only one feature vector forms the basis for these models, facilitating accurate and focused Prediction. This approach represents the innovative aspect of the hybrid models, as it employs advanced techniques and hyperparameter tuning to outperform traditional deep learning methods in forecasting tasks.

6. Analysis of Results

6.1. Quantitative Analysis

Table 10 presents an in-depth evaluation of the performance of various DL models on three different kinds of sunspot data: smoothed SSN data for 13 months, yearly SSN data, and mean monthly SSN data. The models used here include conventional deep-learning architectures such as LSTM, GRU, CNN, RNN, and BiLSTM, as well as newly proposed hybrid models, including hybrid1, hybrid2, hybrid3, hybrid4, and a hybrid ensemble model. The effectiveness of these models is evaluated using four main metrics, namely RMSE, MAE, MSE, and R-squared. The dataset-wise analysis is given below:

- Performance on 13-Smoothed Monthly SSN Data:** Traditional models perform at different levels on the monthly SSN data smoothed by 13. The LSTM model shows strong performance with an RMSE of 4.598, an MAE of 1.965, and an R^2 score of 0.9953, indicating a reliable predictive accuracy. The GRU model is similar but has a slightly higher RMSE of 4.726 and a slightly lower R^2 of 0.9951. As far as the models are concerned, the CNN performs slightly better, with the lowest value of RMSE at 4.465 and the highest value of R^2 at 0.9956, indicating excellent predictive power of the model. The RNN and BiLSTM models also performed poorly because their RMSE values were the highest,

Table 10 Performance Comparison of Deep-Learning Models for Sunspot Number Prediction.

Dataset	Model	RMSE	MAE	MSE	R-squared
13-smoothed Monthly SSN	<i>Traditional Models</i>				
	LSTM	4.598	1.965	21.184	0.995353
	GRU	4.726	1.737	22.411	0.995085
	CNN	4.465	1.755	19.973	0.995619
	RNN	4.899	1.544	24.393	0.994650
	BiLSTM	5.041	2.493	25.531	0.994400
	<i>Proposed Models</i>				
	Hybrid1	4.118	1.420	16.980	0.996276
	Hybrid2	4.186	1.465	17.529	0.996155
	Hybrid3	4.242	1.361	18.011	0.996050
	Hybrid4	4.225	1.403	17.873	0.996080
	Hybrid Ensemble	4.063	1.339	16.509	0.996379
Yearly SSN	<i>Traditional Models</i>				
	LSTM	23.771	18.259	566.141	0.874971
	GRU	24.814	19.163	618.025	0.863513
	CNN	23.310	17.685	543.472	0.879978
	RNN	23.985	17.875	576.401	0.872705
	BiLSTM	26.642	19.992	711.623	0.842843
	<i>Proposed Models</i>				
	Hybrid1	23.594	17.963	556.952	0.877001
	Hybrid2	22.651	17.192	513.555	0.886585
	Hybrid3	22.622	17.323	512.122	0.886900
	Hybrid4	23.105	17.701	534.182	0.882028
	Hybrid Ensemble	22.110	16.904	488.940	0.892020
Monthly Mean SSN	<i>Traditional Models</i>				
	LSTM	24.753	17.911	612.771	0.881144
	GRU	25.085	18.217	629.351	0.877928
	CNN	24.757	17.946	612.961	0.881107
	RNN	24.829	17.986	616.531	0.880414
	BiLSTM	24.743	17.925	612.322	0.881232
	<i>Proposed Models</i>				
	Hybrid1	24.697	17.912	610.034	0.881676
	Hybrid2	24.868	18.027	618.529	0.880027
	Hybrid3	24.921	18.033	621.171	0.879514
	Hybrid4	24.691	17.932	609.702	0.881738
	Hybrid Ensemble	24.606	17.853	605.520	0.882550

while their R^2 values were the lowest, indicating lesser accuracy. When compared, the hybrid models suggested usually do better than the traditional ones. Among them, Hybrid1 shows enhanced prediction accuracy with an RMSE of 4.117 and an R^2 of 0.9963. Hybrid2, Hybrid3, and Hybrid4 perform better than conventional models, showing lower RMSE and MAE values. Significantly, the Hybrid Ensemble model performed best in all

performance measures with the lowest RMSE at 4.062, an MAE of 1.338, and the highest R^2 at 0.9964. This outstanding performance highlights the effectiveness of combining several hybrid models to improve predictive accuracy.

- Performance on Yearly SSN Data:** The traditional models have higher error rates when tested on annual SSN data compared to their performance on monthly data. The LSTM model has an RMSE of 23.77 and an R^2 of 0.8750, while the CNN has a slightly higher performance with an RMSE of 23.31 and an R^2 of 0.8800. The GRU and RNN models show higher RMSE and lower R^2 values, meaning less accurate predictions. The BiLSTM model shows the worst performance with an RMSE of 26.64 and an R^2 of 0.8428. On the other hand, the proposed hybrid models show better performance than these traditional models. Hybrid2 and Hybrid3 have the lowest RMSE values, with Hybrid2 achieving an RMSE of 22.65 and an R^2 of 0.8866, while Hybrid3 has a slightly lower RMSE of 22.62 and an R^2 of 0.8869. The Hybrid Ensemble model accurately predicts yearly SSN data with the lowest RMSE of 22.11, MAE of 16.90, and the highest R^2 of 0.8920. It shows the vast benefits of hybrid and ensemble models in improving the accuracy of predictions compared to traditional methods.
- Performance on Monthly Mean SSN Data:** Traditional models like LSTM, CNN, and BiLSTM perform similarly on monthly mean SSN data, with RMSE values between 24.74 and 24.75 and R^2 scores around 0.8812. The GRU and RNN models, although slightly less precise, still show results in a similar range. Generally, the proposed hybrid models provide higher accuracy than traditional models. Hybrid4 has the lowest RMSE of 24.69, with other hybrid models closely behind. Nonetheless, the Hybrid Ensemble model distinguishes itself by achieving the lowest RMSE of 24.61, MAE of 17.85, and the highest R^2 of 0.8826. The outstanding performance of this model highlights how hybrid and ensemble modelling techniques can improve forecast accuracy for monthly mean SSN data. Overall, the table shows that the suggested hybrid models consistently outperform conventional deep-learning models across all types of sunspot data. The Hybrid Ensemble model provides the most precise forecasts, exhibits minor error metrics, and achieves the highest R^2 scores. This comparative analysis emphasizes the significant benefits of combining hybrid and ensemble methods to maximize predictive accuracy in challenging datasets like sunspot activity.

The comparative analysis shown in Table 11 points out the better performance of proposed hybrid models compared to state-of-the-art methods using various sunspot datasets. The metrics of performance, i.e., RMSE, MAE, MSE, and R^2 , clearly show that the proposed models outperform the models in forecasting accuracy consistently. For the 13-smoothed monthly sunspot number time series dataset, the existing top-performing model, sConvLSTM1D (Kumar and Kumar 2023a,b), achieves an RMSE of 13.15, while the Hybrid Ensemble Model lowers the RMSE to 4.062 substantially. Likewise, the MAE of the hybrid ensemble model is 1.338, which is considerably less than that of CNN-BiGRU (8.16). Additionally, the R^2 value of the hybrid ensemble model is 0.996379, significantly improving upon the top state-of-the-art highest R^2 value of 0.885. These findings reveal that the suggested models are more accurate and reliable predictions.

For the annual sunspot number dataset, the suggested models also result in a marked improvement in the prediction accuracy. The current top-performing method, Stacked LSTM (Elgamal 2020), has an RMSE of 23.478, which is reduced to 22.110 by the proposed hybrid ensemble model. The MAE also goes down from 18.99 (PHM-MAX-ARIMA-ETS SVM) to 16.904, which shows that the proposed model generates lower absolute errors in predictions. The R^2 score of 0.89202 is also higher than other models, strengthening the efficacy of the hybrid ensemble method in identifying sunspot trends. For the dataset of monthly

Table 11 Comparative Analysis of Performance Metrics Across Different Datasets and Models.

Dataset	Reference	Approach	RMSE	MAE	MSE	R ²
13-smoothed Monthly SSN	<i>State-of-the-art Models</i>					
	Pala and Atici (2019)	6-partLSTM	35.9	–	–	–
	Panigrahi et al. (2021)	PHM-MAX-ARIMA	22.73	–	–	–
	Lee (2020)	EMD+LSTM	30	–	–	–
	Ramadevi and Bingi (2022)	LSTM	10.85	–	–	–
	Kumar and Kumar (2024)	CNN-BiGRU	18.84	8.16	–	–
	Moustafa and Khodairy (2023)	LSTM-ARIMA	17.76	–	–	0.93
	Kumar and Kumar (2023b)	sConvLSTM1D	13.15	–	–	0.885
	<i>Proposed Models</i>					
	This article	Proposed hybrid1	4.117	1.419	16.979	0.996276
	This article	Proposed hybrid2	4.185	1.465	17.528	0.996155
	This article	Proposed hybrid3	4.241	1.360	18.010	0.99605
	This article	Proposed hybrid4	4.225	1.403	17.872	0.99608
	This article	Proposed Hybrid Ensemble	4.062	1.338	16.508	0.996379
Yearly SSN	<i>State-of-the-art Models</i>					
	Elgamal (2020)	Stacked LSTM	23.478	–	–	–
	Panigrahi et al. (2021)	PHM-MAX-ARIMA-ETS SVM	25.46	18.99	–	–
	<i>Proposed Models</i>					
	This article	Proposed hybrid1	23.594	17.962	556.95	0.877001
	This article	Proposed hybrid2	22.651	17.192	513.55	0.886585
	This article	Proposed hybrid3	22.622	17.323	512.12	0.8869
	This article	Proposed hybrid4	23.105	17.701	534.18	0.882028
	This article	Proposed Hybrid Ensemble	22.110	16.904	488.94	0.89202
	<i>State-of-the-art Models</i>					
Monthly Mean SSN	Lee (2020)	EMD+LSTM	[30,60]	–	–	–
	Jiang et al. (2023)	LSTM-NN	50.4	–	–	–
	<i>Proposed Models</i>					
	This article	Proposed hybrid1	24.697	17.91	610.03	0.881676
	This article	Proposed hybrid2	24.868	18.02	618.53	0.880027
	This article	Proposed hybrid3	24.921	18.03	621.17	0.879514
	This article	Proposed hybrid4	24.691	17.93	609.70	0.881738
	This article	Proposed Hybrid Ensemble	24.606	17.85	605.52	0.88255

Note: Bold values indicate results from the proposed models in this article. Dashes (–) indicate unavailable metrics

mean sunspot number, the RMSE range for the state-of-the-art model (Lee's EMD+LSTM 2020) is 30–60, whereas that of the suggested hybrid ensemble model is significantly less at 24.606. The MAE is also lowered to 17.85, indicating the proposed method's reliability in minimizing forecasting errors. Also, the R^2 value of 0.88255 is the maximum for all the models, supporting that the devised hybrid ensemble model learns complex temporal patterns in sunspot activity more efficiently.

The overall results supported the idea that proposed hybrid models, especially the hybrid ensemble model, are superior to other techniques in all the datasets. Lower RMSE and MAE values indicate that the models suggested are more accurate in their predictions, and higher R^2 values verify their performance of capturing the intrinsic trends in sunspot activity better. All these improvements owe their occurrence to incorporating deep learning processes that effectively capture short-term fluctuations and long-term trends. Thereafter, the intended hybrid models outline a significant contribution to sunspot number forecasting, providing a sounder and precise forecasting model instead of former established methods.

6.2. Graphical Analysis

The graphical analysis has been divided into three part: 1) Prediction pattern analysis using line plots, 2) correlation analysis using scatter plots and Performance distribution analysis using Box plots.

6.2.1. Prediction Pattern Analysis Using Line Plots

Figures 8, 9, and 10 present line plots comparing predictions from both proposed and traditional deep-learning models. The plots are organized in a matrix format, with rows representing traditional model variations and columns showing proposed hybrid models (Hybrid 1, 2, 3, 4, and ensemble models). Each plot incorporates the actual test dataset, traditional deep learning model predictions, and proposed hybrid model predictions.

- In Figure 8, the proposed ensemble model demonstrates the closest alignment with the original SSN series compared to both hybrid proposed models and traditional deep-learning models. While the hybrid proposed models show superior effectiveness over traditional models, GRU and RNN exhibit particularly low correlation with the original test dataset pattern. Among traditional models, LSTM demonstrates better performance in sunspot forecasting.
- Analysis of Figure 9 reveals that the ensemble hybrid model achieves the closest pattern match to the SSN series test dataset, followed by other proposed hybrid models. Although the hybrid proposed models generally show superior performance, the LSTM model exhibits relatively stronger pattern alignment with the original test dataset compared to other traditional models. The RNN model shows the largest deviation from the original pattern among all models.
- Figure 10 demonstrates that both the proposed hybrid ensemble model and hybrid models maintain closer alignment with the original SSN test data compared to traditional deep learning models. A notable observation is the tendency for the Proposed Hybrid Ensemble and other models to show increasing divergence at higher Time values. The ensemble approach proves particularly effective for longer sequence predictions. While LSTM and GRU models exhibit notably higher error rates, the proposed models maintain better stability over time. Similarly, both LSTM and Bi-LSTM show greater deviation from the actual test dataset compared to the proposed models.



Figure 8 Sub-plot of line plot of traditional vs proposed models concerning actual test data of 13-Smoothed Monthly sunspot dataset.

6.2.2. Predictive Correlation Analysis Using Scatter Plots

Figures 11, 12, and 13 present scatter plots comparing the actual versus forecasted values for both traditional and proposed models, including the hybrid ensemble model. The plots are arranged in a matrix format where rows represent traditional models and columns show proposed models, with a reference line indicating the ideal correlation.

- In Figure 11, the proposed models demonstrate consistently higher correlations and better alignment with the reference line compared to traditional models. The hybrid ensemble model exhibits the strongest correlation among all models. While LSTM and GRU show relatively weaker correlations, RNN and BiLSTM demonstrate notably stronger correlations among the traditional models.

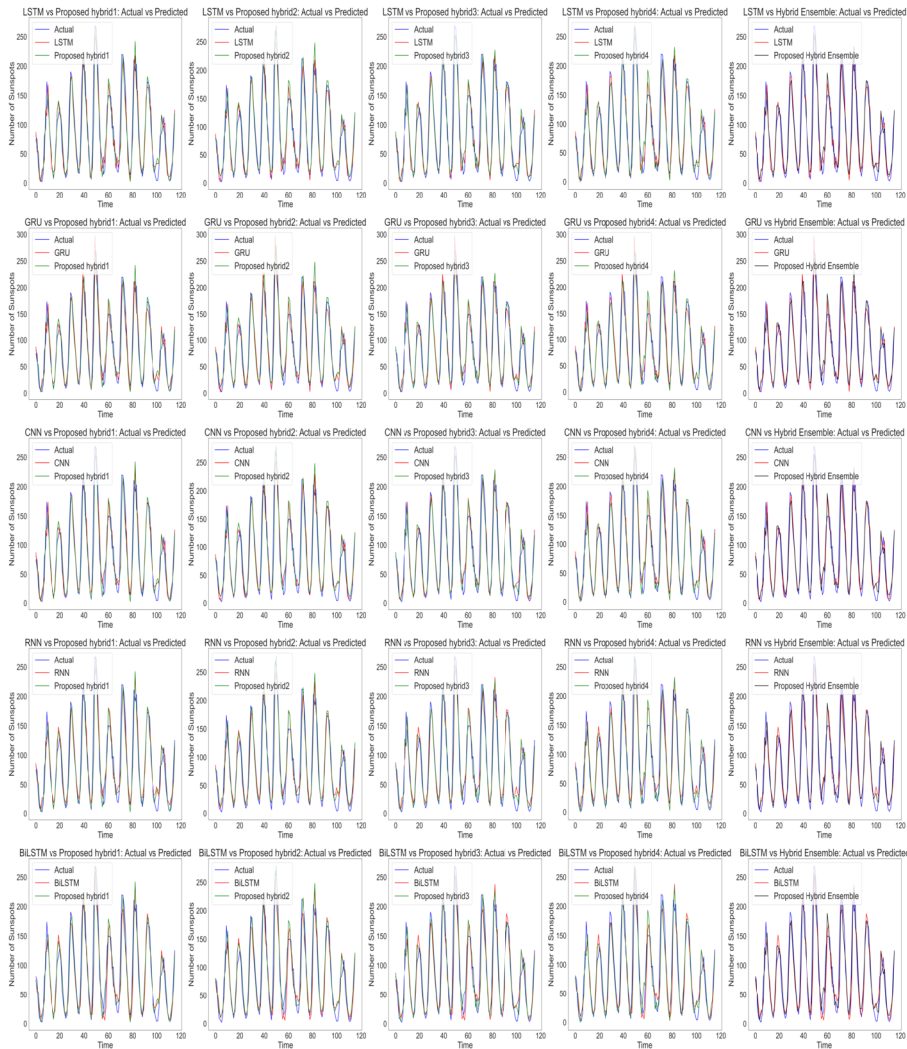


Figure 9 Sub-plot of line plot of traditional vs proposed models concerning actual test data Yearly sunspot dataset.

- The analysis of Figure 13 reveals similar patterns. LSTM and GRU models show the weakest correlations between their predictions and the test dataset compared to other models. Among traditional models, RNN and BiLSTM show more promising results, though their performance remains below that of the proposed models. The hybrid ensemble model maintains the highest correlation across all models, including other proposed hybrid models.
- Figure 12 further confirms these findings, showing that the proposed hybrid ensemble model achieves the highest correlation and closest alignment with the reference line compared to both hybrid and traditional deep-learning models. Consistent with the 13-monthly and Monthly Mean datasets, LSTM and GRU demonstrate the poorest corre-

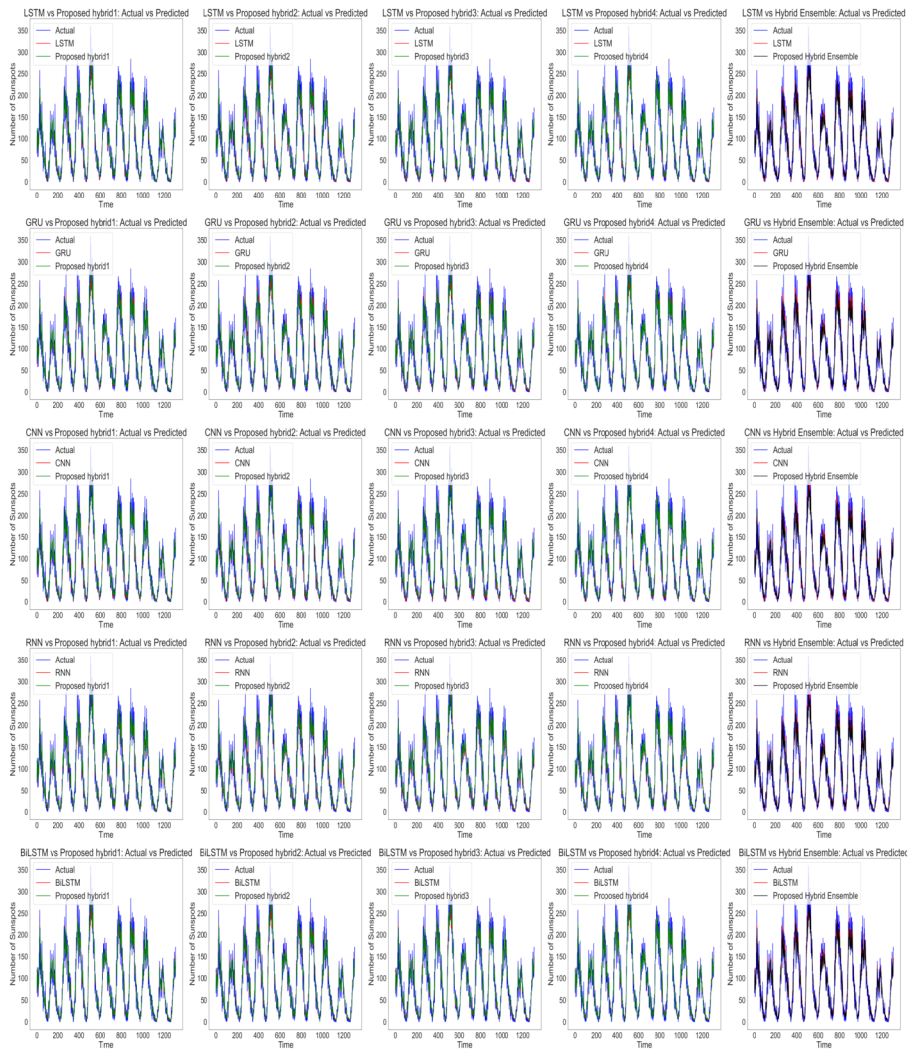


Figure 10 Sub-plot of line plot of traditional vs proposed models concerning actual test data Monthly Mean Sunspot dataset.

lation among all models, while BiLSTM and RNN show improved correlation among traditional models, though still not matching the performance of the proposed models.

6.2.3. Performance Distribution Analysis Using Box Plots

All figures (14, 15, and 16) use consistent color coding: Traditional models are represented in “sky blue” and Hybrid models are displayed in “light green”. The box plots show performance data from multiple runs of 20 unique runs for 13-month smoothed data, Yearly sunspot dataset and Monthly mean dataset. The performance distribution has been analysis based on the performance measures as RMSE, MSE and R^2 which are mention as given below:



Figure 11 Sub-Scatter plot of traditional vs proposed models of 13-Smoothed Monthly Sunspot data.

- RMSE Performance Analysis:** Hybrid1 achieves an RMSE of 4.11, while the hybrid ensemble model (combining hybrid2 and hybrid4) achieves a superior RMSE of 4.06, demonstrating the best performance among all models. In Yearly Sunspot dataset, Hybrid3 shows the best individual performance with an RMSE of 22.62, but the hybrid ensemble model surpasses this with an RMSE of 22.11. The hybrid ensemble model maintains its superiority with an RMSE of 22.11, outperforming both traditional and individual hybrid models over the Yearly Sunspot dataset. In Monthly Mean dataset, The hybrid ensemble model maintains its superiority with an RMSE of 22.11, outperforming both traditional and individual hybrid models.
- MAE Performance Analysis:** The hybrid ensemble model demonstrates the lowest MAE values compared to both traditional and other hybrid models over 13-Month Smoothed

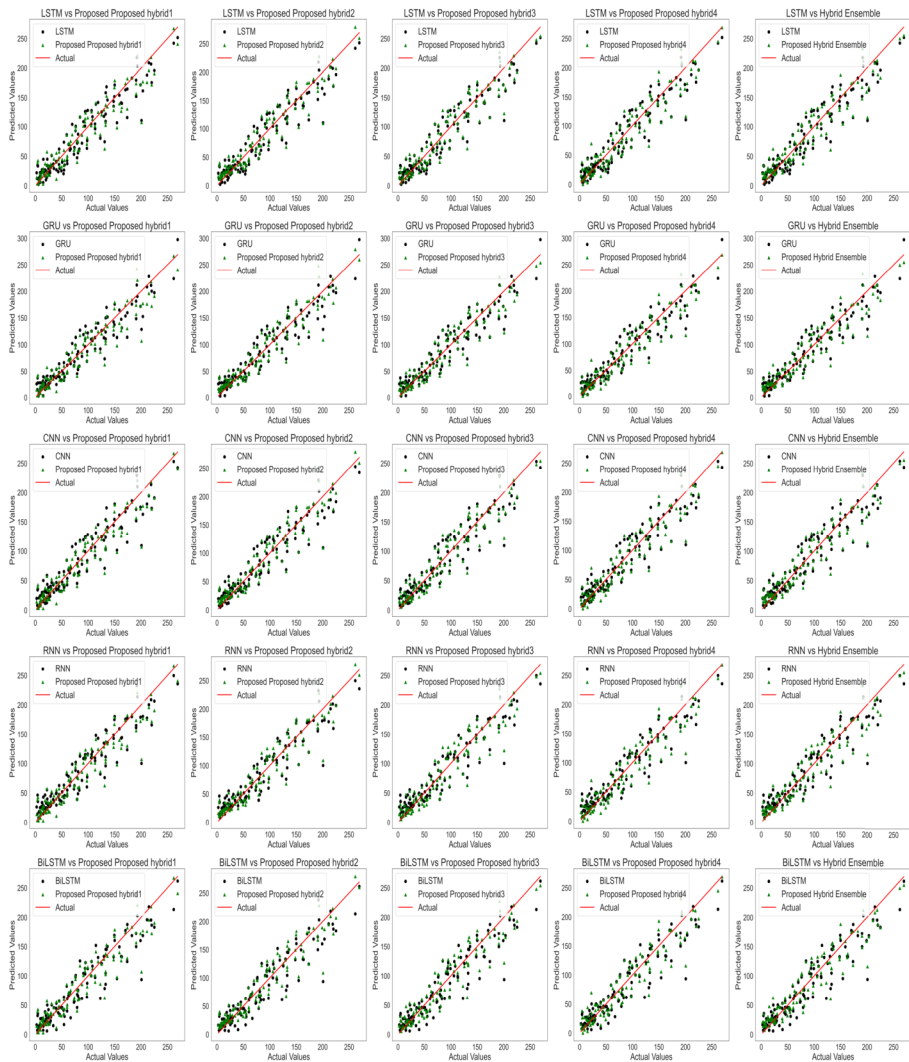


Figure 12 Sub-Scatter plot of traditional vs proposed models of yearly sunspot data.

Monthly dataset. All hybrid models show superior performance compared to traditional models, with the hybrid ensemble model achieving the lowest MAE values over Yearly Sunspot series. In Monthly Mean dataset, The hybrid ensemble model consistently outperforms all other models in MAE measurements, showing better predictive accuracy.

- MSE Performance Analysis:** The hybrid ensemble model shows the best performance in MSE metrics compared to all other models over 13-Month Smoothed Monthly dataset. Similarly, The hybrid ensemble model maintains the lowest MSE values, demonstrating superior error minimization using Yearly Sunspot Data. In Monthly Mean dataset, The consistent with other datasets, the hybrid ensemble model achieves the lowest MSE values among all models.

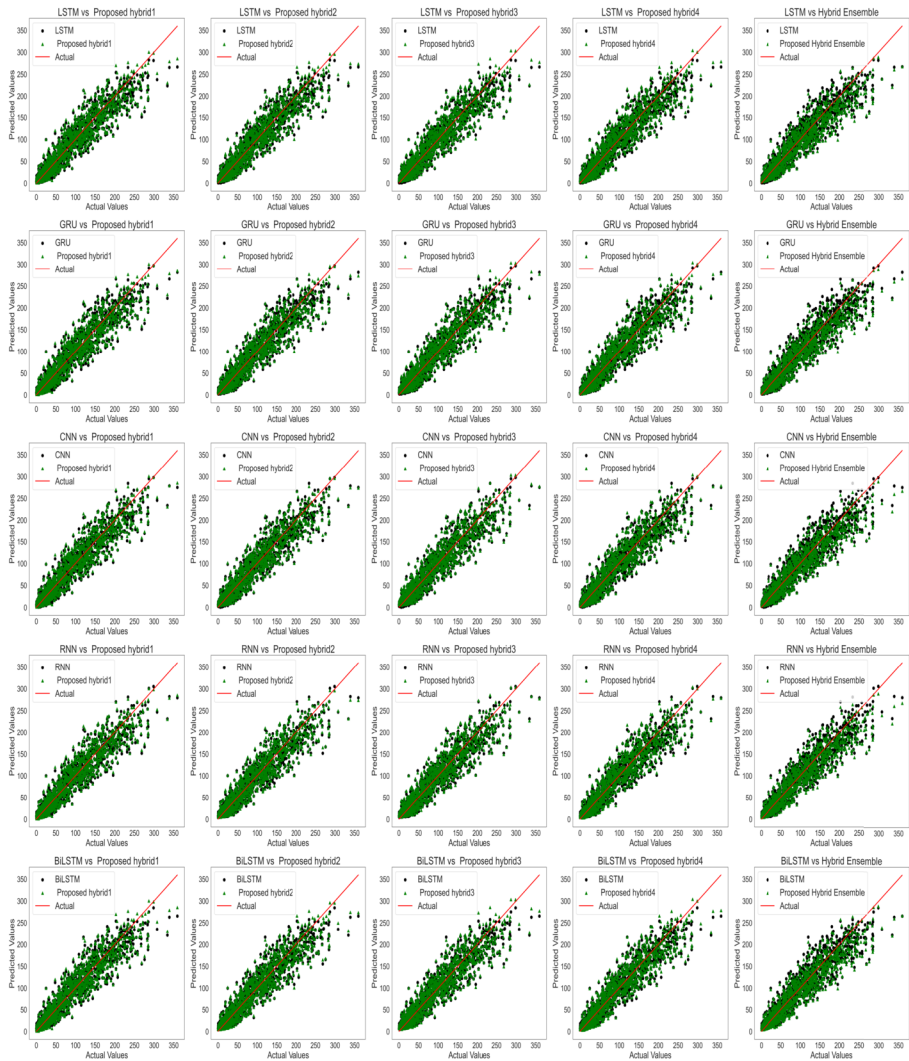


Figure 13 Sub-Scatter plot of traditional vs proposed models of Monthly Mean Sunspot data.

- R^2 Performance Analysis:** In 13-Month Smoothed Monthly dataset, The hybrid ensemble model achieves a high R^2 value of 0.994650, confirming excellent data alignment. The hybrid ensemble model demonstrates robust performance with an R^2 value of 0.8920 over Yearly Sunspot dataset. In Monthly Mean dataset, The model maintains its strong performance with an R^2 value of 0.8920, validating its predictive capability.

The hybrid ensemble model, combining hybrid2 and hybrid4, consistently demonstrates superior performance across all datasets and metrics, highlighting the effectiveness of the ensemble approach in handling complex sunspot activity data.

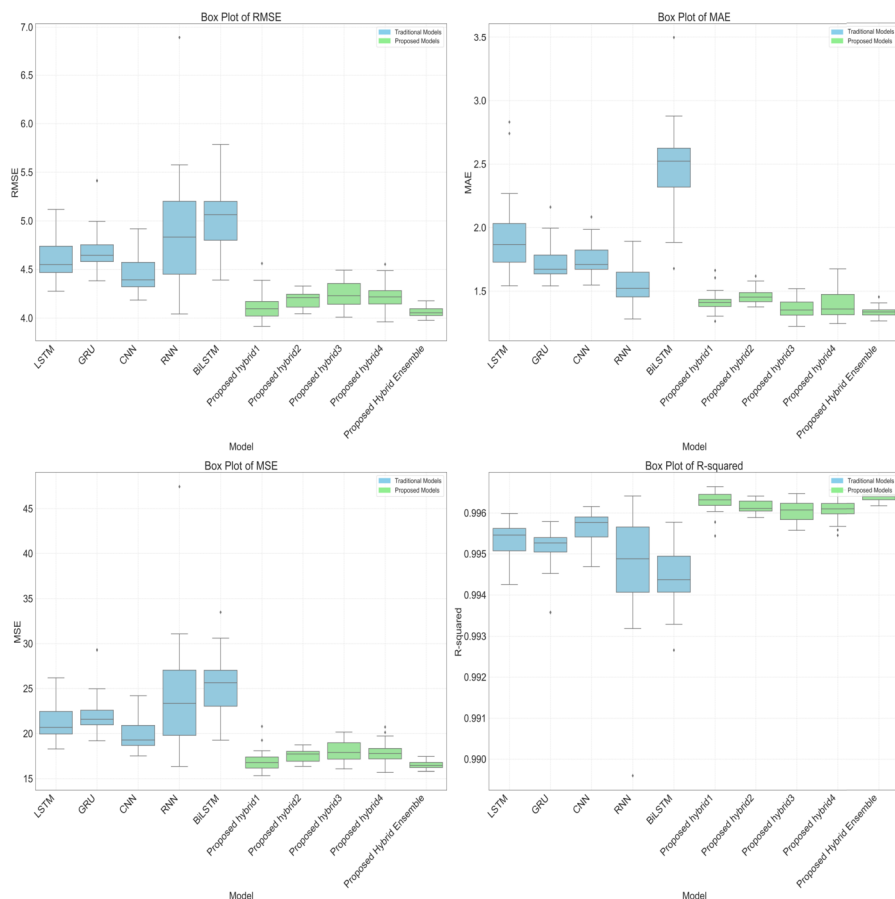


Figure 14 Sub-box plot of proposed hybrid models of 13-Smoothed Monthly Sunspot data.

6.3. Non-parametric Statistical Test Using Friedman Ranking

Table 12 presents the Mean Friedman rankings, providing non-parametric statistical comparisons of model performance across multiple datasets. The Friedman test, used to determine treatment differences across testing rounds, evaluates performance by considering average rankings of different models. The analysis reveals that the Proposed Hybrid Ensemble model achieves the best performance with an average rank of 1, demonstrating statistical significance with a z-value of 2.101947149 and a p-value of 0.035557906. However, when applying Holm's correction for multiple comparisons, the adjusted p-value (0.355579057) suggests potential type I error concerns that may affect the significance level. Traditional deep-learning models demonstrate relatively higher rankings, indicating inferior performance. The LSTM model shows an average rank of 6.333333333 with a z-value of -0.389249472 , and its performance lacks statistical significance (p-value: 0.69709161). Similarly, the GRU model, with an average rank of 9 and z-value of -1.634847783 , shows a trend toward significance (p-value: 0.102080961) but fails to exceed the adjusted threshold (Holm-corrected p-value: 0.91872865). The proposed hybrid models consistently outperform conventional approaches. Proposed Hybrid1 achieves an average rank of 3.666666667 (z-value:

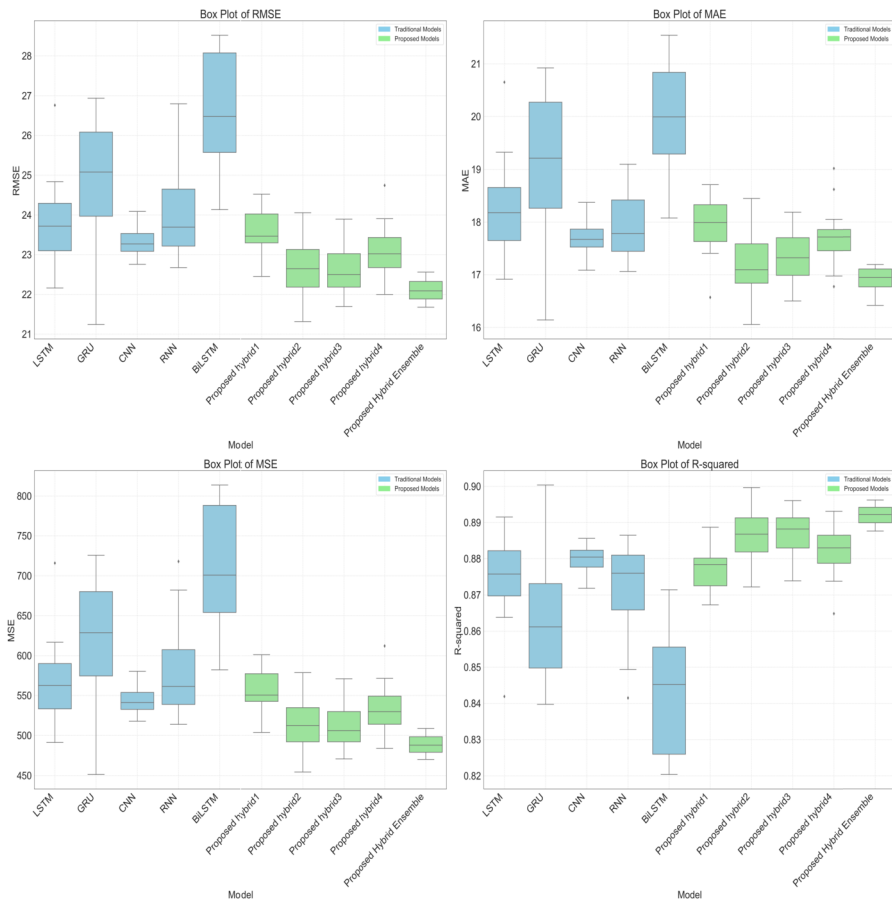


Figure 15 Sub-box plot of proposed hybrid models of yearly sunspot data.

0.856348839, p-value: 0.391804855, not significant), while Proposed Hybrid4 shows similar strong performance with an average rank of 3.333333333 (z-value: 1.012048627, p-value: 0.311514804, not significant). Overall, the results demonstrate that hybrid models, particularly the Proposed Hybrid Ensemble, exhibit superior predictive accuracy compared to pure deep-learning models in sunspot data analysis. The implementation of both the Friedman test and Holm's correction ensures robust statistical comparison, highlighting the enhanced precision and reliability that hybrid approaches bring to complex time series prediction tasks.

7. Solar Cycle 26 Forecast Peak Values

Figure 17 presents a comprehensive analysis of sunspot numbers (SSN) that combines historical data with future predictions, offering valuable insights into Solar Cycle 26. The visualization spans more than three centuries, starting from 1700 and extending into the future until 2040, revealing the dynamic nature of solar activity patterns. The historical record,

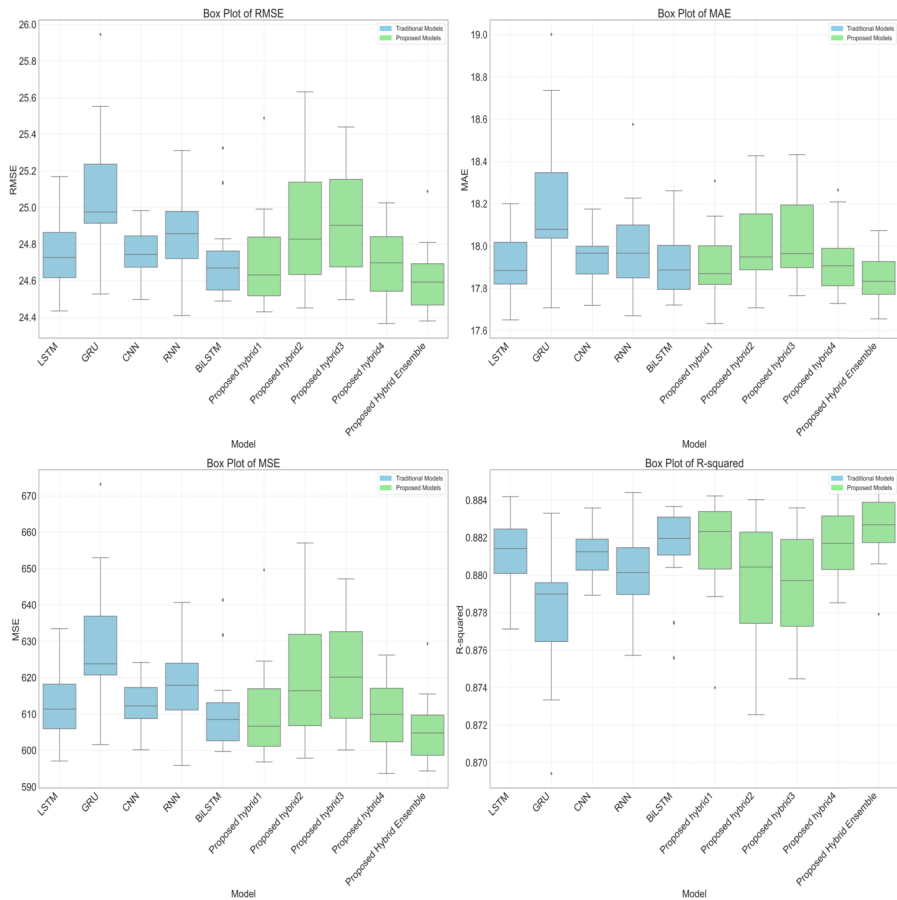


Figure 16 Sub-box plot of proposed Monthly Mean Sunspot data hybrid models.

Table 12 Mean Friedman rankings of the algorithms, z-value, p-value, and Holm-corrected p-value for Various Models.

Model	Average rank	z-value	p-value	Holm-corrected p-value
LSTM	6.3333	0.3892	0.6970	1
GRU	9	1.6348	0.1020	0.9187
CNN	5.6666	0.0778	0.9379	1
RNN	8	1.1677	0.2429	1
BiLSTM	8	1.1677	0.2429	1
Proposed hybrid1	3.666	0.8563	0.3918	1
Proposed hybrid2	4.6666	0.3892	0.6970	1
Proposed hybrid3	5.3333	0.0778	0.9379	1
Proposed hybrid4	3.3333	1.0120	0.3115	1
Proposed Hybrid Ensemble (hybrid2+hybrid4)	1	2.1019	0.0355	0.3555

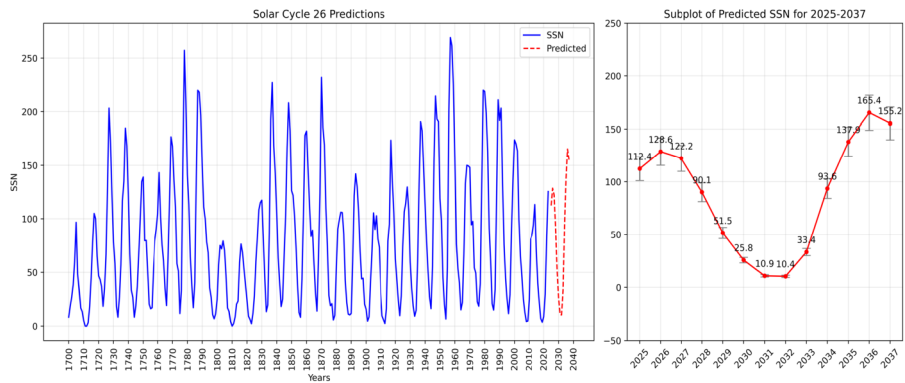


Figure 17 Yearly Forecasting of the SSN.

represented by a solid blue line, demonstrates the characteristic 11-year solar cycle, showing regular fluctuations in solar activity. These cyclical patterns have been consistently observed throughout the recorded history of solar observations, with each cycle displaying unique characteristics in terms of intensity and duration. This historical context provides a crucial foundation for understanding future predictions. Looking toward the future, the forecast is depicted by a dashed red line spanning from 2025 to 2037. The prediction indicates a gradual increase in solar activity following a recent period of minimum activity. A detailed examination of the right panel reveals specific predicted values across this timeframe: the cycle begins with an SSN of 112.39 in early 2025, reaches its lowest point of 10.41 in 2032, and then climbs to a peak of 165.35 in 2036. The forecast concludes with a slight decline to 155.25 in 2037, suggesting the beginning of the cycle's declining phase. These predictions carry significant implications for various fields of study and practical applications. The forecasted solar activity patterns provide essential information for understanding and preparing for space weather events, potential climate impacts, and technological vulnerabilities. The accuracy of these predictions becomes particularly crucial for developing appropriate mitigation strategies to protect Earth's environment and technological infrastructure from solar-induced disturbances. This underscores the importance of continued refinement in SSN forecasting methods to enhance our preparedness for future solar activity cycles.

Table 13 presents a comparative summary of predicted peak times and amplitudes for Solar Cycle 25 (SC25) and Solar Cycle 26 (SC26) from various studies employing different forecasting methodologies. These approaches range from statistical and dynamo-based models to machine learning techniques such as LSTM, RNN, and hybrid deep-learning models. The diversity in predicted results stems primarily from differences in data preprocessing, input feature selection, model architecture, and forecasting philosophy (e.g., precursor-based vs. physics-based vs. data-driven methods). As a result, both the forecasted peak amplitude and timing vary across studies. Moreover, in entries like “SC26: 2036” and “SC26: 165.35,” the former denotes the predicted year of peak solar activity for Solar Cycle 26, while the latter indicates the predicted peak sunspot number. The inclusion of SC26 predictions by some authors adds a long-term perspective and allows for cross-cycle trend comparisons. Our proposed CNN-GRU-RNN hybrid model forecasts the SC26 peak in the year 2036, with a predicted peak sunspot number of 165.35, aligning with high-activity projections in recent literature.

Table 13 A comparative analysis of sunspot predicted peaks and solar cycle durations from various literature sources about the proposed methods.

References	Approach	Associated time	Forecasted peak
Kakad, Kakad, and Ramesh (2017)	Shannon Entropy Technique	SC25: 2024	63 ± 11.3
Singh and Bhargawa (2017)	Hurst Exponent	SC25: June 2024	102.8 ± 24.6
Li et al. (2018)	Statistical Forecasting Technique	SC25: October 2024	168.5 ± 16.3
Pesnell and Schatten (2018)	Precursor Technique	SC25: 2025.2 ± 1.5	135 ± 25
Petrovay et al. (2018)	Precursor Technique	SC25: 2024	130
Upton and Hathaway (2018)	Dynamo-based Forecasting	-	110
Han and Yin (2019)	Statistical Forecasting	SC25: 2023	228.8 ± 40.5
Labonville, Charbonneau, and Lemerle (2019)	Dynamo-based Forecasting	SC25: $2025 + 0.89$	$89 + 24-14$
Pala and Atici (2019)	LSTM	SC25: October 2024	137.8 ± 31.3
Sello (2019)	Nonlinear Prediction Technique	SC25: July 2023 ± 1	107 ± 10
Benson et al. (2020)	DNN	SC25: May 2025 ± 1	106 ± 19.75
Du (2020)	Statistical Forecasting	SC25: 2023.2 ± 1.1	167.3
Goncalves, Echer, and Frigo (2020)	Gaussian Process Regression	SC25: 2024	117
Courtillot, Lopes, and Le Mouél (2021)	Neural Network	SC25: 2023	57 ± 17
Guo, Jiang, and Wang (2021)	Dynamo-based Forecasting	-	126
Wang, Li, and Guo (2021b)	LSTM	SC25: 2023	114.3
Xiong et al. (2021)	Precursor Technique	SC25: March 2024 ± 1	140.2
Du (2022)	Correlation Statistics	SC25: December 2024 ± 0.5	135.5 ± 33.2
Nghiem et al. (2022b)	Hilbert Transform	SC25: August 2023 ± 0.16	229 ± 76
Prasad et al. (2022)	LSTM	SC25: July 2025 ± 0.16	171 ± 3.4
Zhu, Zhu, and He (2022)	LSTM	SC25: 2024	143.6 ± 8.7
Kumar and Kumar (2023b)	Stacked ConvLSTM1D	SC25: January 2025	140.84
Zhu et al. (2023)	LSTM	SC25: February 2024	213
Su et al. (2023)	N-BEATS	SC25: 2024	133.9 ± 7.2
Liu et al. (2023)	LSTM	SC26: 2035	SC25: 135.2, SC26: 135.0
Espuña Fontcuberta et al. (2023)	RNN	SC25: 2024	131 ± 14
Prasad, Roy, and Sarkar (2024)	Double-layered LSTM	SC25: July 2023 + 3 months	56 ± 5
Kumar and Kumar (2024)	CNN-BiGRU, GRC	SC25: 2024	145.535, 143.641
Su et al. (2024)	SCINet	SC25: January 2024	127.59 ± 10.03
Xu, Jain, and Xing (2024)	RF	SC25: 2025	SC25: 154, SC26: 141
Kumar (2024)	vD2R2w	SC25: 2025	SC25: 127.16 ± 6.83 , SC26: 191.71 ± 43.37

Table 13 (Continued)

References	Approach	Associated time	Forecasted peak
Cao et al. (2024a)	Informer, FWin-Informer	SC25: April 2025, SC25: April 2025	120.69, 129.45
Cao et al. (2024b)	Transformer, Informer	SC25: Mar 2025, SC25: May 2025	104.6, 178.5
Cao et al. (2025)	Informer, FWin-Informer	SC25: January 2027, SC25: October 2026	3.46e-6, 3.88e-6
This Article	CNN-GRU-RNN Hybrid	SC26: 2036	SC26: 165.35

8. Conclusion

This research evaluates deep-learning models' performance in sunspot activity prediction, comparing traditional methods with hybrid and ensemble approaches. Traditional models (LSTM, GRU, CNN, RNN, and BiLSTM) show varying effectiveness across different datasets - CNN excels with 13-month smoothed monthly data, while LSTM and CNN perform well with annual data. However, these conventional models demonstrate limitations with monthly average SSN data, indicating challenges in capturing sunspot activity's full complexity. Hybrid models (hybrid1 through hybrid4) consistently outperform conventional approaches across all datasets by combining multiple neural network configurations. This integration leverages individual model strengths, significantly improving prediction accuracy. The Hybrid Ensemble model emerges as the superior performer, achieving the lowest error metrics and highest R^2 scores, demonstrating exceptional capability in handling sunspot prediction complexities. Applied to the 26th solar cycle (2025–2037), the Hybrid Ensemble model forecasts specific SSN values: 112.39 in 2025, peaking at 165.35 in 2036, before decreasing to 155.25 in 2037. These predictions align with current solar cycle observations, accurately capturing sunspot activity's cyclical nature. The study demonstrates hybrid and ensemble modeling's advantages over traditional methods. The Hybrid Ensemble model's superior performance highlights the importance of these techniques for improving prediction accuracy in complex, dynamic datasets like sunspot activity. This research advances predictive analytics in solar activity forecasting, suggesting opportunities for further model refinement and application to more complex prediction challenges.

Acknowledgments The authors acknowledge the SILSO World Data Centre operated by the Royal Observatory of Belgium, SIDC, for providing sunspot number (SSN) databases used in this work. The datasets were downloaded on 14 September 2024 from the SILSO website (www.sidc.be/SILSO/home) WDC-SILSO, Royal Observatory of Belgium (2025). The datasets were well-maintained and supplied by the SIDC to support the analysis and results of this study.

Author Contributions Aman Kumar: Conceptualization, Methodology, Software, Formal analysis, Data Curation, Writing - Original Draft, Visualization, Vipin Kumar: Methodology, Validation, Formal analysis, Investigation, Resources, Writing - Review & Editing, Supervision,.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

References

- Abdulkadir, R.A., Hasan, M.K., Islam, S., Gadekallu, T.R., Pandey, B., Safie, N., et al.: 2023, *Earth Sci. Inform.* **16**, 3931. [DOI](#).
- Benson, B., Pan, W.D., Prasad, A., Gary, G.A., Hu, Q.: 2020, Forecasting Solar Cycle 25 using deep neural networks. *Sol. Phys.* **295**(5), 65.
- Cao, J., Xu, T., Deng, L., Zhou, X., Li, S., Liu, Y., Wang, W., Zhou, W.: 2024b, An improved prediction of Solar Cycles 25 and 26 using the informer model: Gnevyshev peaks and North–South asymmetry. *Astrophys. J.* **969**(2), 120.
- Cao, J., Xu, T., Deng, L., Zhou, X., Zhao, X., Xiang, N., Li, F., Wan, M., Zhou, W.: 2024a, Hemispheric prediction of solar cycles 25 and 26 from multivariate sunspot time-series data via Informer models. *RAS Tech. Instrum.*
- Cao, J., Xu, T., Deng, L., Wan, M., Zhao, X., Peng, Y., Zhou, X., Wang, W., Huang, L., Wang, W., et al.: 2025, Multi-source forecast of solar cycle flare activity using the novel informer-based models. *Space Weather* **23**(4), e2024SW004322.
- Chola, C., Benifa, J.V.B.: 2022, *Glob. Transit. Proc.* **3**, 177. [DOI](#).
- Chung, J., Gulcehre, C., Cho, K., Bengio, Y.: 2014, arXiv preprint. [arXiv](#).
- Courtillot, V., Lopes, F., Le Mouél, J.L.: 2021, On the prediction of solar cycles. *Sol. Phys.* **296**, 1.
- Dang, Y., Chen, Z., Li, H., Shu, H.: 2022, A comparative study of non-deep learning, deep learning, and ensemble learning methods for sunspot number prediction. *Appl. Artif. Intell.* **36**(1), 2074129.
- Dhuri, D.B., Bhattacharjee, S., Hanasoge, S.M., Kiran Mahapatra, S.: 2022, *Astrophys. J.* **939**, 64. [DOI](#).
- Du, Z.L.: 2020, The solar cycle: predicting the peak of Solar Cycle 25. *Astrophys. Space Sci.* **365**(6), 104.
- Du, Z.: 2022, Predicting the maximum amplitude of Solar Cycle 25 using the early value of the rising phase. *Sol. Phys.* **297**(5), 61.
- Elgamal, M.: 2020, Sunspot time series forecasting using deep learning. *Int. J. Comput. Inf. Technol.* **9**(2).
- Espuña Fontcuberta, A., Ghosh, A., Chatterjee, S., Mitra, D., Nandy, D.: 2023, Forecasting Solar Cycle 25 with physical model-validated recurrent neural networks. *Sol. Phys.* **298**(1), 8. [DOI](#).
- Goncalves, I.G., Echer, E., Frigo, E.: 2020, Sunspot cycle prediction using warped Gaussian process regression. *Adv. Space Res.* **65**(1), 677.
- Guo, W., Jiang, J., Wang, J.X.: 2021, A dynamo-based prediction of Solar Cycle 25. *Sol. Phys.* **296**, 1.
- Han, Y.B., Yin, Z.Q.: 2019, A decline phase modeling for the prediction of Solar Cycle 25. *Sol. Phys.* **294**(8), 107. [DOI](#).
- Hasoon, S.O., Al-Hashimi, M.M.: 2022, Hybrid deep neural network and long short term memory network for predicting of sunspot time series. *Comput. Sci.* **17**, 955.
- Hu, A., Rodriguez, A.: 2023, A Hybrid CNN-LSTM Model for Predicting Solar Cycle 25. *J. Coll. Stud. Ret.* **12**(1).
- Jiang, J., Wang, J.-X., Jiao, Q.-R., Cao, J.-B.: 2018, *Astrophys. J.* **863**, 159. [DOI](#).
- Jiang, Y., Xiong, Y., et al.: 2023, Prediction of sunspot activity cycle based on Long Short-Term Memory (LSTM) network models. *Acad. J. Comput. Inf. Sci.* **6**(13), 178. Francis Academic Press.
- Kakad, B., Kakad, A., Ramesh, D.S.: 2017, Shannon entropy-based prediction of Solar Cycle 25. *Sol. Phys.* **292**(7), 95. [DOI](#)
- Kumar, V.: 2024, Enhancing Solar Cycle 25 and 26 forecasting with vipin-deep-decomposed-recomposed rolling-window (vD2R2w) model on sunspot number observations. *Sol. Phys.* **299**(10), 147.
- Kumar, A., Kumar, V.: 2023a, Decompose and recomposed time series data-based ensemble model (DRE-TS) for effective temperature prediction. In: *Proc. 2nd Int. Conf. on Automation, Computing and Renewable Systems (ICACRS)*, IEEE, 1694.
- Kumar, A., Kumar, V.: 2023b, Stacked 1D Convolutional LSTM (sConvLSTM1D) model for effective prediction of sunspot time series. *Sol. Phys.* **298**, 121. [DOI](#).
- Kumar, A., Kumar, V.: 2024, Forecast of Solar Cycle 25 based on hybrid CNN-Bidirectional-GRU (CNN-BiGRU) model and novel Gradient Residual Correction (GRC) technique. *Adv. Space Res.* **73**(8), 4342. [DOI](#). Elsevier.
- Kumar, B., Yadav, N., et al.: 2023, A novel hybrid model combining β SARMA and LSTM for time series forecasting. *Appl. Soft Comput.* **134**, 110019.
- Labonville, F., Charbonneau, P., Lemerle, A.: 2019, A dynamo-based forecast of Solar Cycle 25. *Sol. Phys.* **294**(6), 82.
- Lee, T.: 2020, EMD and LSTM hybrid deep learning model for predicting sunspot number time series with a cyclic pattern. *Sol. Phys.* **295**(6), 82. Springer.
- Li, F.Y., Kong, D.F., Xie, J.L., Xiang, N.B., Xu, J.C.: 2018, Solar cycle characteristics and their application in the prediction of Cycle 25. *J. Atmos. Sol.-Terr. Phys.* **181**, 110. [DOI](#).
- Li, Q., Wan, M., Zeng, S.-G., Zheng, S., Deng, L.-H.: 2021, Predicting the 25th solar cycle using deep learning methods based on sunspot area data. *Res. Astron. Astrophys.* **21**, 184. [DOI](#).

- Liu, X., Zeng, S., Deng, L., Zeng, X., Zheng, S.: 2023, Predicting the 25th and 26th solar cycles using the long short-term memory method. *Publ. Astron. Soc. Jpn.* **75**(3), 691.
- Moustafa, S.S.R., Khodairy, S.S.: 2023, Comparison of different predictive models and their effectiveness in sunspot number prediction. *Phys. Scr.* **98**(4), 045022.
- Nghiem, T.-L., Le, V.-D., Le, T.-L., Maréchal, P., Delahaye, D., Vidosavljevic, A.: 2022a, *Proc. 2022 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, 1. DOI.
- Nghiem, T.-L., Le, V.-D., Le, T.-L., Maréchal, P., Delahaye, D., Vidosavljevic, A.: 2022b, Applying Bayesian inference in a hybrid CNN-LSTM model for time-series prediction. In: *Proc. 2022 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, IEEE, 1.
- Pala, Z., Atici, R.: 2019, Forecasting sunspot time series using deep learning methods. *Sol. Phys.* **294**(5), 50. DOI.
- Panigrahi, S., Pattanayak, R.M., Sethy, P.K., Behera, S.K.: 2021, Forecasting of sunspot time series using a hybridization of ARIMA, ETS and SVM methods. *Sol. Phys.* **296**(1), 6. DOI.
- Pesnell, W.D.: 2016, An early prediction of the amplitude of Solar Cycle 25. *Space Weather* **14**, 10. DOI.
- Pesnell, W.D., Schatten, K.H.: 2018, An early prediction of the amplitude of Solar Cycle 25. *Sol. Phys.* **293**, 112.
- Petrovay, K.: 2020, Solar cycle prediction. *Living Rev. Sol. Phys.* **17**(1), 2.
- Petrovay, K., Nagy, M., Gerják, T., Juhász, L.: 2018, Precursors of an upcoming solar cycle at high latitudes from coronal green line data. *J. Atmos. Sol.-Terr. Phys.* **176**, 15.
- Prasad, A., Roy, S., Sarkar, A.: 2024, Hemispheric prediction of Solar Cycle 25 based on a deep learning technique. *Adv. Space Res.* **73**(3), 2119.
- Prasad, A., Roy, S., Sarkar, A., Panja, S.C., Patra, S.N.: 2022, Prediction of Solar Cycle 25 using deep learning based long short-term memory forecasting technique. *Adv. Space Res.* **69**, 798. DOI.
- Prasad, A., Roy, S., Sarkar, A., Panja, S.C., Patra, S.N.: 2023, *Sol. Phys.* **298**, 50. DOI.
- Ramadevi, B., Bingi, K.: 2022, Time series forecasting model for sunspot number. In: *Proc. 2022 International Conference on Intelligent Controller and Computing for Smart Power (ICICCSPP)*, IEEE, 1.
- Sello, S.: 2019, Solar cycle activity: an early prediction for Cycle#25. arXiv preprint. arXiv.
- Singh, A.K., Bhargawa, A.: 2017, An early prediction of 25th solar cycle using Hurst exponent. *Astrophys. Space Sci.* **362**(11), 199.
- Su, X., Liang, B., Feng, S., Dai, W., Yang, Y.: 2023, Solar Cycle 25 prediction using N-BEATS. *Astrophys. J.* **947**(2), 50.
- Su, X., Liang, B., Feng, S., Cai, Y., Dai, W., Yang, Y.: 2024, Solar cycle prediction using a combinatorial deep learning model. *Mon. Not. R. Astron. Soc.* **527**(3), 5675.
- Upton, L.A., Hathaway, D.H.: 2018, An updated Solar Cycle 25 prediction with AFT: the modern minimum. *Geophys. Res. Lett.* **45**(16), 8091.
- Wang, Q.-J., Li, J.-C., Guo, L.-Q.: 2021a, A dynamo-based prediction of Solar Cycle 25. *Res. Astron. Astrophys.* **21**, 012. DOI.
- Wang, Q.J., Li, J.C., Guo, L.Q.: 2021b, Solar cycle prediction using an extended short-term memory deep learning model. *Res. Astron. Astrophys.* **21**(1), 012.
- Waykar, S.B., Kale, A.W., Karlekar, N.P.: 2022, *Deep Learning-Based Solar Activity Prediction Using Sunspot Number and Solar Radio Flux, Advances in Parallel Computing Algorithms, Tools and Paradigms*, IOS Press, Amsterdam, 40.
- WDC-SILSO, Royal Observatory of Belgium: 2025, Sunspot Number Data Archive. Accessed: January 2025. Available at <https://www.sidc.be/SILSO/datafiles>.
- Xiong, Y., Lu, J., Zhao, K., Sun, M., Gao, Y.: 2021, Forecasting Solar Cycle 25 using comprehensive precursor combination and multiple regression techniques. *Mon. Not. R. Astron. Soc.* **505**(1), 1046.
- Xu, Q., Jain, R., Xing, W.: 2024, Data-driven forecasting of sunspot cycles: pros and cons of a hybrid approach. *Sol. Phys.* **299**(2), 25. DOI.
- Yang, J., Liu, S., Xuan, S., Chen, H.: 2024, A hybrid model based on CEEMDAN-GRU and error compensation for predicting sunspot numbers. *Electronics* **13**(10), 1904. DOI.
- Zhu, H., Zhu, W., He, M.: 2022, Solar Cycle 25 prediction using an optimized long short-term memory mode with F10. 7. *Sol. Phys.* **297**, 157. DOI.
- Zhu, H., Chen, H., Zhu, W., He, M.: 2023, Predicting Solar Cycle 25 using an optimized extended short-term memory model based on sunspot area data. *Adv. Space Res.* **71**(8), 3521.

Authors and Affiliations

Aman Kumar¹ · Vipin Kumar²

✉ V. Kumar
rt.vipink@gmail.com

A. Kumar
amankumardce2000@gmail.com

¹ Department of Computer Science (Artificial Intelligence), Meerut Institute of Engineering & Technology, Meerut, Uttar Pradesh, 250005, India

² Department of Computer Science and Information Technology, Mahatma Gandhi Central University, Motihari, Bihar, 845401, India